

Population-Genetics-Informed Neural Networks for a Single-Population Coalescent Under Selection

Peter Beerli¹

¹Department of Scientific Computing, Florida State University

September 30, 2026

Abstract

Coalescent likelihood methods are rigorous but slow. Neural networks trained on simulated data are fast, but treat the simulation as a black box, disconnected from theory. We introduce PGINN (Population-Genetics-Informed Neural Networks), which adds a Kullback–Leibler term to an ordinary data-fitting loss. The term compares the coalescent rate under predicted parameters to the rate under true parameters, following physics-informed neural networks. We test it on a single population under directional selection, a deterministic hard sweep. We use two regimes resembling a single-locus scan: 10 loci pooled within a candidate-region summary or one locus, at sample sizes of $n=10$ and $n=20$ individuals. Every result is replicated across 10 independent runs, with paired significance tests.

In all four regimes PGINN recovers the selection strength s better than the mean square error (MSE) loss alone, in every one of the 10 replicates (paired $p \leq 1.3 \times 10^{-4}$), and removes most of MSE’s downward bias for strong selection; the price is a small loss of accuracy for the population-scaled mutation rate θ . For detecting selection outright, a simple classical statistic (how star-like the genealogy looks) is as good as either network at 10 loci; with a single genealogy, the networks detect strong sweeps the classical statistic misses, and combining the two in a logistic regression does best. External validation with MSPRIME, a different simulator, shows that the network recovers the expected hitchhiking footprint and no spurious peak in neutral data. Applied to the *Adh* locus of *Drosophila melanogaster* (176 lines), the network’s estimate of θ agrees with a Bayesian coalescent estimate, and none of the 15 non-recombining segments show evidence of a hard sweep. Adding the coalescent term to the loss thus improves estimates of the selection strength s , the parameter that the term constrains directly, while estimates of the population-scaled mutation rate θ become slightly less accurate.

Keywords: selective sweep; coalescent theory; physics-informed neural networks; population-genetics-informed neural networks; simulation-based inference; population genomics

1 Introduction

Two paradigms dominate demographic and selective inference from genetic data. Full-likelihood coalescent methods integrate over the space of genealogies that could have produced the data

(Kingman, 1982; Felsenstein et al., 1999), giving uncertainty quantification but scaling poorly to genome-wide data or complex models. Simulation-based and machine-learning approaches instead train a model, often a neural network, to map simulated data directly to parameters (Cranmer et al., 2020; Schrider and Kern, 2018); once trained, inference is fast, but the network has no explicit connection to the population-genetic theory that generated its training data.

In the physical sciences, neural Networks face the same disconnect: a trained network is fast, but ignores the governing differential equations that describe the system it is modeling. Raissi et al. (2019) introduced physics-informed neural networks (PINN) to address this disconnect by embedding those differential equations directly into the training loss, so that a network’s predictions remain consistent with known physical laws even where data are sparse or noisy. This paper develops the population-genetics analog: **PGINN** (Population Genetics-Informed Neural Networks), in which a Kullback–Leibler (KL) divergence (Kullback and Leibler, 1951) between the coalescent distribution implied by a network’s predicted parameters and the coalescent distribution used to simulate the training data is mixed into the loss alongside ordinary supervised error.

We study a single population under directional selection, where the coalescent-rate term used in the loss (Section 2.4) is itself only an approximation to the true selective genealogy process, and ask whether a theory-informed loss still helps under that approximation. We keep the model small and the selection mechanism simple: a deterministic hard-sweep rescaling of the coalescent rate (Section 2), in the tradition of classical coalescent approximations for a completed selective sweep (Kaplan et al., 1989; Barton, 1998; Braverman et al., 1995), so that the question stays about the loss function, not about building a new simulator. We make four contributions. (i) A self-contained derivation, not a new theoretical result, connecting the large, population-scaled selection parameter used in sweep models to the familiar $[0, 1]$ -bounded per-generation selection coefficient, and a discussion of what a hard-sweep model omits relative to a soft sweep (Section 2). (ii) A side-by-side comparison of the PGINN and MSE (mean squared error) losses over data regimes chosen to resemble a single-locus selection scan (Section 3.4): 10 loci pooled within a candidate-region summary vs. a single locus, each at two sample sizes, $n=10$ and $n=20$ individuals (Section 4), replicated 10 times per regime to separate a genuine loss-function effect from training noise. (iii) A practically motivated extension of that comparison to a detection problem (distinguishing a neutral locus from a selected one, as one would want to do when scanning many candidate regions) rather than only estimating s given that selection is present, benchmarked against a simple classical (non-network) detection statistic (Section 4.4). (iv) An external validation of the trained network on an independent simulator (msprime) with recombination and a neutral (no-sweep) control, which the network was never trained on (Section 4.6; Kelleher et al., 2016). All code, a custom coalescent simulator generating a Newick tree and then transforming the node timings and tree pattern into Kendall–Colijn tree-summary features (Kendall and Colijn, 2016; Khodaei et al., 2023, 2025), and a plain MLP (multilayer perceptron) regressor architecture, is self-contained and part of the PGINN repository (<https://github.com/pbeerli/pginn>).

2 Model

72

2.1 From a per-generation selection coefficient to a population-scaled parameter

73

In standard population-genetic usage, a selection coefficient s_g is bounded, $s_g \in [0, 1]$: under genic (additive) selection the favored allele has relative fitness $1 + s_g$ against 1 for the wild type ($s_g=0$: neutral; $s_g=1$: the wild type contributes no offspring at all, the maximal possible selective advantage). Under a discrete-generation Wright–Fisher model, the favored allele’s frequency p updates as

74
75
76
77
78

$$p' = \frac{p(1 + s_g)}{1 + s_g p}, \quad (1)$$

79

which, for weak selection per generation, gives the diffusion approximation

80

$$\frac{dp}{d\tau} = s_g p(1 - p), \quad \tau = \text{time in generations.} \quad (2)$$

81

Coalescent time is rescaled in units of a population-size constant c : the same constant relating θ to the per-generation mutation rate, for example, $\theta = c\mu$ with $c=2N$ for haploid populations, $t=\tau/c$. N as the number of individuals in a Wright–Fisher population; for more general populations, we may write effective population N_e . Substituting into Eq. 2,

82
83
84
85

$$\frac{dp}{dt} = c s_g p(1 - p) \equiv s p(1 - p), \quad s \equiv c s_g, \quad (3)$$

86

so s , the *population-scaled* selection coefficient, directly analogous to $\theta=c\mu$ rather than s_g itself, is what actually enters the coalescent-time-scaled dynamics. Because c is large (population size) and hard sweeps are of interest for weak or strong per-generation s_g and small or large population sizes, s ranges from about 1 to the thousands, not over $[0, 1]$. Throughout the rest of this paper, and in the accompanying code, we use s for the population-scaled selection coefficient. This s should not be confused with the bounded, per-generation s_g of Eq. 1.

87
88
89
90
91
92

$\theta=c\mu$ and $s=c s_g$ are both population-scaled, dimensionless compound parameters, folding population size (c) into a per-generation rate (μ or s_g); s in Eq. 4–6 below is this compound quantity, not a per-generation rate.

93
94
95

Genealogies alone cannot distinguish a large population with weak mutation from a small population with strong mutation; only the product θ is identifiable. The same is true of selection: only the product $s=c s_g$ is identifiable, not s_g and c separately. This is exactly why the hitchhiking and sweep literature reports s (often written $2Ns_g$) rather than s_g (Barton, 1998).

96
97
98
99

2.2 Backward-time hard-sweep rescaling of the coalescent

100

Solving Eq. 3 for a hard sweep and expressing the allele-frequency trajectory backward from the present ($t=0$) gives

101
102

$$x(t; s) = \frac{x_0}{x_0 + (1 - x_0) e^{st}}, \quad x_0 = 0.999, \quad (4)$$

103

where x_0 is the present-day frequency of the selected allele, fixed here close to 1. This backward-time form is obtained by substituting $\tau=T-t$ into the forward solution and taking the hard-sweep

104
105

limit $p_0 \rightarrow 0$. At $s=0$, $x(t; s)=x_0 \approx 1$ for all t ; for $s > 0$, $x(t; s)$ decreases toward 0 as t grows, compressing coalescence into a burst near the sweep’s origin. Genealogies, whether simulated or estimated from sequence data, measure time not in these coalescent units but in expected mutations per site, $u=\mu\tau=\theta t$, the scale of branch lengths; on this scale a neutral genealogy is about θ tall. In terms of u , the mutation-scaled size of the subpopulation carrying the selected allele, θ_e , and the coalescent rate for k lineages become

$$\theta_e(u; \theta, s)=\theta x(u/\theta; s), \quad \lambda(u, k)=\frac{k(k-1)}{\theta_e(u; \theta, s)}. \quad (5)$$

On the u scale, the sweep trajectory therefore runs at rate $s/\theta=s_g/\mu$: the same s collapses x within a shorter span of branch length when θ is small, which is why θ and s have to be estimated jointly. This uses the same time-varying-effective-population-size rescaling technique as the coalescent with exponential growth (Hudson, 1990) and approximations of a hitchhiking sweep (Kaplan et al., 1989; Barton, 1998), though the trajectory itself differs (exponential for growth, the sweep-specific collapse of Eq. 4 here), but in both cases confining the sample’s ancestors to an ever-shrinking pool compresses genealogies toward a near-star shape as the respective parameter (growth rate, or s) grows.

2.3 Hard sweeps vs. soft sweeps

The $p_0 \rightarrow 0$ approximation in Section 2.2 is the defining feature of a *hard sweep*: the beneficial allele arises as a single new mutation, so every sampled copy must, looking backward, funnel through that one founding lineage before the allele frequency could have been any higher than $1/c$. A *soft sweep* (Hermisson and Pennings, 2005) arises instead when the beneficial allele either recurs independently by mutation multiple times, or was already segregating as standing genetic variation at an appreciable frequency $x_{sv} \gg 1/c$ before an environmental shift made it favored. In either case, several independent lineages carry the allele at the onset of selection, so coalescence does not have to funnel through a single ancestor: genealogies remain comparably less star-like than a hard sweep of the same strength and completion time, because the extra lineages can persist, uncoalesced, back through the entire selected phase.

Our model, with $x(t; s)$ continuing to shrink toward the numerical floor $x_{\min}=10^{-4}$ used in the simulator’s discretization (Section 3) rather than leveling off at some $x_{sv} > 0$, is a pure hard-sweep approximation. A soft-sweep extension is a small change: replace $x_{\min}$ with a free “standing-frequency” parameter x_{sv} , and once $x(t)$ reaches it, switch to a neutral coalescent among the $k_0 \approx c x_{sv}$ founding lineages (constant θ from that point backward) rather than continuing to shrink θ_e further, representing the standing-variation phase that preceded selection. We do not implement or evaluate this here; it is an incremental special case worth adding later. All results below use this hard-sweep approximation, not selection in general.

2.4 The PGINN loss

For a genealogy of n samples there are $K=n-1$ internal-node ages $u_K > \dots > u_2$ (on the branch-length scale of Eq. 5, sorted descending: u_K is the root, u_2 the most recent coalescence), with $k=2, \dots, K+1$ lineages present at each. The KL divergence between the sweep-rescaled coalescent

processes implied by predicted vs. true (θ, s) , evaluated at these observed ages, is

$$\mathcal{L}_{\text{PGINN}} = \frac{1}{K} \sum_{k=2}^{K+1} \left[\log \frac{\hat{\lambda}_k}{\lambda_k} + \frac{\lambda_k}{\hat{\lambda}_k} - 1 \right], \quad \hat{\lambda}_k = \frac{k(k-1)}{\hat{\theta} x(u_k/\hat{\theta}; \hat{s})}, \quad \lambda_k = \frac{k(k-1)}{\theta_{\text{true}} x(u_k/\theta_{\text{true}}; s_{\text{true}})}, \quad (6)$$

the closed-form KL divergence between two exponential distributions per interval (this reduces to the standard single-population neutral coalescent KL loss when $s=0$ up to the fixed constant $x_0=0.999$, since $x(t;0)=x_0$ is then a rescaling of θ , not literally θ itself).

θ here is anchored at the present ($t=0$): since $x(0;s)=x_0$ for every s (Section 2.2), $\theta_e(0)=\theta x_0 \approx \theta$, the same role θ_0 plays in the coalescent with exponential growth (Section 2.2), where a present-day θ_0 anchors an exponentially rescaled $\theta_e(t)$ in exactly the same way. Unlike growth, though, the population’s total size c never changes in this model; only the fraction of that constant population carrying the beneficial allele, $x(t;s)$, shrinks backward in time. Equivalently, the sample’s ancestors are confined to that shrinking derived-allele subpopulation, while the complementary, still-wild-type fraction $1-x(t;s)$, not ancestral to the sample, grows backward in time as $x(t;s) \rightarrow 0$: from the sample’s own lineages’ point of view this looks exactly like a shrinking effective population size, even though the population as a whole never changes size.

The total training objective is the fixed-ratio mixture $\mathcal{L}=(1-w)\mathcal{L}_{\text{MSE}}+w\mathcal{L}_{\text{PGINN}}$; an adaptive mixture, in which w itself is learned or allowed to vary with the informativeness of the data, is a natural extension we return to in Section 5.

This is a theoretically shaped supervised loss, not an unsupervised physics constraint. Unlike a PINN’s governing-equation residual, which is typically evaluated against the network’s own output and known physics without needing a ground-truth label, $\mathcal{L}_{\text{PGINN}}$ ’s reference rate λ_k in Eq. 6 is computed from the true $(\theta_{\text{true}}, s_{\text{true}})$, the same labels $\mathcal{L}_{\text{MSE}}$ already uses.

3 Methods

3.1 The coalescent-with-hard-sweep simulator

For our training and evaluations, we use genealogies as data. We create our training genealogies using the coalescent-based simulator `simtree.py` (Khodaei et al., 2025). In the single-population, constant- θ neutral ($s=0$) model, it draws a coalescent rate $k(k-1)/\theta$ from the current number of lineages k and θ , draws the next coalescence time from the corresponding exponential distribution, and repeats until one lineage remains, following the standard Kingman coalescent.

Genealogies under hard sweep conditions are approximated by changing θ_e (Eq. 5), not θ itself, at prescribed backward times, recording $(u_j, \theta_{e,j})$ pairs. Throughout, θ is the single constant drawn once per replicate (Section 3.2), and $\theta_{e,j}=\theta x(u_j/\theta; s)$ (Eq. 5) is its time-dependent rescaling by the sweep trajectory; only θ_e , never θ , changes across breakpoints. The same technique, a piecewise-constant, time-varying coalescent parameter, is used for arbitrary step-function population-size histories (Griffiths and Tavaré, 1994; Hudson, 2002). This approach makes θ_e , not θ , piecewise-constant in backward time.

The simulator’s event loop draws a candidate next-event time under the *current* θ_e ; if that

candidate time overshoots the next scheduled breakpoint u_j , the drawn event is discarded, backward time is advanced exactly to u_j , θ_e is updated to $\theta_{e,j}$, and the loop repeats.

3.2 Simulation

We simulated haploid samples of size $n=10$, $n=20$, or $n=176$ (Section 4.8), with θ drawn log-uniformly on $[0.001, 0.05]$. For training and validation data, $s=0$ with probability 0.25 and otherwise s is drawn log-uniformly on $[1, 1000]$, so that neutral genealogies are well represented and weak and strong sweeps receive equal weight on the log scale; the sweep prediction sets draw s log-uniformly on $[1, 1000]$ and the neutral prediction sets fix $s=0$. For $s > 0$ we build a piecewise-constant approximation to $\theta_e(t; \theta, s) = \theta x(t; s)$ (Eq. 4) and feed it to the time-varying- θ mechanism of Section 3.1: we use 40 breakpoints $x_1 > x_2 > \dots > x_{40}$, evenly spaced in $\log x$ from just below x_0 (starting at $0.999 x_0$, since $x(0; s) = x_0$ exactly and starting the grid at x_0 itself would place the first breakpoint at $t=0$) down to a numerical floor $x_{\min}=10^{-4}$, each mapped to a backward time by inverting Eq. 4 and converting to the branch-length scale, $u=\theta t$,

$$t(x; s) = \frac{1}{s} \ln \left(\frac{x_0(1-x)}{x(1-x_0)} \right), \quad (7)$$

giving the timepoints file $(\theta t(x_i; s), \theta x_i)_{i=1}^{40}$ that the simulator reads directly (Section 3.1).

We space these breakpoints in x rather than in t because the collapse of $x(t; s)$ happens on a timescale set entirely by s . Setting $x=1/2$ in Eq. 7 gives the transition's midpoint,

$$t_{\text{mid}} = \frac{1}{s} \ln \left(\frac{x_0}{1-x_0} \right) = \frac{\ln(999)}{s}, \quad (8)$$

since $x_0=0.999$ makes $x_0/(1-x_0)=999$. A fixed, time-spaced grid would need to know already t_{mid} (and hence s) to place points near it. Grid points chosen evenly in $\log x$ avoid this: each lands at whatever backward time the trajectory actually reaches that frequency, whatever s turns out to be, so the same 40 breakpoints always straddle t_{mid} , whether $s=4$ puts it at $u=\theta t_{\text{mid}} \approx 0.035$ or $s=400$ at $u \approx 0.00035$ (both at $\theta=0.02$).

The choice of 40 breakpoints is arbitrary, not derived from a formal discretization-error bound. Table 1 is the empirical check that it suffices.

The star ratio is a diagnostic we construct for this calibration: the fraction of the TMRCA (time to the most recent common ancestor) not already covered by the second-deepest split, where a small value means a star-like tree. At a fixed $\theta=0.02$ and $n=10$, the star ratio drops from 0.45–0.47 at $s \leq 4$ (indistinguishable from neutral) through 0.25 at $s=20$ to 0.06 at $s=2000$, confirming that this mechanism produces the expected transition from Kingman-like to star-like genealogies (Kaplan et al., 1989) as s increases.

The within-tree coefficient of variation (CV) is a second, per-tree star-likeness diagnostic, independent of the across-replicate TMRCA statistics: the standard deviation (SD) of a genealogy's own K internal-node ages divided by their mean, averaged across replicates. This CV falls in step with the star ratio, and the TMRCA variance collapses by three orders of magnitude, the classic sweep signature of reduced genetic diversity (Maynard Smith and Haigh, 1974; Braverman et al.,

Table 1: Calibration of the sweep-rescaled simulator at $\theta=0.02$, 200 replicate genealogies per s with sample sizes of 10 and 20, respectively.

Sample size	s	TMRCa mean	TMRCa std	star ratio	within-tree CV
10	0	0.0168	0.010	0.450	1.212
	1	0.0185	0.011	0.458	1.241
	4	0.0163	0.0084	0.470	1.200
	20	0.00875	0.0014	0.248	0.948
	100	0.00224	0.00014	0.081	0.544
	400	0.000634	0.000026	0.062	0.314
	2,000	0.000144	0.0000054	0.056	0.163
20	0	0.0193	0.011	0.440	1.621
	1	0.0188	0.010	0.436	1.606
	4	0.0191	0.0094	0.431	1.604
	20	0.00895	0.0012	0.207	1.218
	100	0.00226	0.00012	0.076	0.707
	400	0.000643	0.000030	0.061	0.410
	2,000	0.000144	0.0000056	0.048	0.229

1995). Table 1 also reports the same diagnostic at $n=20$ and confirms the same interpolation holds at the larger sample size used elsewhere in this paper.

Based on this, the prior range $s \in [1, 1000]$ spans “no visible effect” through “strong, clearly star-like sweep” at the θ values used here, so the network sees the whole continuum from weak to strong selection, not just the two extremes, and can learn how genealogy shape varies continuously with s ; drawing s log-uniformly gives each decade of selection strength the same weight. $n=10$ ($K=9$ intervals) is enough to see the transition; Table 1 confirms the same qualitative transition at $n=20$ ($K=19$ intervals), so we use the same s range for every sample size. Because $s_g=s/c$ (Section 2.1), s up to 1000 corresponds to s_g of at most 0.1 for $c \geq 10^4$ and at most 0.001 for $c=10^6$, well inside the $[0, 1]$ range of a per-generation selection coefficient.

3.3 Features and targets

Each training example averages 10 or 1 replicate genealogies (depending on the regime) simulated under the same (θ, s) : the feature vector is the mean of the Kendall–Colijn m , M , and node-age vectors (Kendall and Colijn, 2016) (225-dimensional at $n=10$, 690-dimensional at $n=20$). Because tip labels carry no meaning across independent replicate genealogies, each locus’s pairwise entries are first sorted by age before averaging, so the mean combines the shallowest pairwise coalescence across loci, the second-shallowest across loci, and so on, rather than averaging tip-labeled entries arbitrarily. For the 1-locus regime this reduces to simply that one genealogy’s own sorted vector, with no averaging.

For the PGINN loss, we also export the mean of the sorted internal-node ages ($K=9$ or $K=19$ values, for $n=10$ or $n=20$, $K=175$ at $n=176$; for 1 locus, this is the genealogy’s observed ages). Equation 6 evaluates $x(u_k/\theta; \cdot)$ at these ages. The prediction targets are $[\log \theta, \log(1+s)]$: both parameters span more than an order of magnitude, and $\log(1+s)$ is 0 at the neutral value $s=0$, which a plain $\log s$ could not represent. Predictions are transformed back, $\hat{s}=e^z-1$ for network output z .

3.4 What does a “locus” mean here?

A natural first design would generate many independent loci per training example. This approach is realistic for θ : effective population size and mutation rate act genome-wide, so pooling many independent, unlinked loci to estimate a shared θ is exactly what standard genome-wide diversity estimators do. It is a poor match for s , which is inherently *locus-specific*: a selective sweep affects one genomic region (the selected site and whatever falls within its linked neighborhood before recombination breaks the hitchhiking signal down), not many independent, unlinked loci that all happen to share the same selection coefficient. A genome scan for selection, as in a genome-wide selection scan for a swept locus, realistically has one candidate region, or a handful of tightly linked sites within it, informative about that region’s own s , not many independent replicates of the same selective event.

We therefore restrict the comparisons below to two choices of “how many loci”: 10 loci pooled within a candidate-region summary, and the extreme of a single locus, rather than also including a many-hundred-loci condition that would not correspond to any realistic single-sweep scan.

Separately, we vary the number of individuals sampled per locus, $n=10$ vs. $n=20$, because, unlike the number of loci, sampling more individuals at the same candidate region is a realistic choice available to a study.

3.5 Network and training

We used a standard Multi-Layer Perceptron (MLP) neural network regressor consisting of two hidden layers (256 units each). The model was trained using the AdamW optimizer for 300 epochs, and we kept the network from the epoch with the lowest validation loss.

We compare $w=0$ (MSE only) against PGINN with weight, $w=0.1$, used in every regime; it was chosen from a sweep over $\{0.05, 0.1, 0.2, 0.3\}$ in all four regimes (Appendix A, Table 9). This comparison spans a 2×2 design: 10 loci or 1 locus per training example (Section 3.4), each at $n=10$ or $n=20$ individuals sampled per locus, giving four regimes in total.

The weight w is selected on the 100-example validation split, never on the independently simulated prediction set that the final numbers are reported on, so the reported prediction-set comparisons are not biased by the hyperparameter search itself. Network predictions are bounded to the support of the training prior ($\theta \in [0.001, 0.05]$, $s \in [0, 1000]$): with log links for both parameters, a network output slightly outside the range seen in training would otherwise translate into an arbitrarily large estimate.

Every regime uses 1000 train / 100 validation / 400 independently simulated prediction examples, and every regime/loss combination, including the detection scoring of Section 4.4, is evaluated in 10 replicates (using different random seeds: $\{67, \dots, 76\}$,) to separate a loss-function effect from training-to-training noise; unless noted otherwise, reported point estimates use the first replicate (seed 67) and reported comparisons use the mean and standard deviation across all replicates, together with a paired t -test on the 10 replicate-level differences (the same seeds are used for both losses in every regime, so the design is paired, not independent).

3.6 External validation on an independent simulator

279

Section 4.6 scores the already-trained neural networks on data from msprime’s SweepGenicSelection model (Baumdicker et al., 2022), a genuinely different (stochastic, recombination-aware) simulator never used in training, under two distinct designs answering two different questions.

The first design checks a capability question (does the learned (θ, s) -to-genealogy relationship generalize at all?) under generous replication: a 500 kb chromosome with a hard sweep fixed at its midpoint is sliced into 100 windows, and each window’s feature vector is averaged over 30 independent replicate simulations of the same sweep, exactly the 10-loci mean convention of Section 3.3, scored with the 10-loci model. msprime reports branch lengths in generations; multiplying them by a per-generation mutation rate μ puts them on the branch-length scale u of Eq. 5. We simulate haploid samples ($c=2N_e$, $N_e=1000$) and choose $\mu=10^{-5}$, i.e. $\theta=c\mu=0.02$, a value in the middle of the training prior; this single constant applies to every window and every replicate, so it fixes the units without being fitted to any feature of the simulated genealogies.

The second design instead asks a single-dataset detection question: Can a candidate sweep site still be localized with only one chromosome and no cross-replicate averaging at all, the situation a genome scan actually faces? Here, a single simulated chromosome is scored tree by tree rather than on a fixed grid of window positions: an arbitrary grid finer than the chromosome’s own recombination density repeats each local tree’s score across several redundant positions instead of adding information. Every one of the tree sequence’s own distinct local trees is instead scored exactly once, with the 1-locus model (matching the true one-genealogy-no-averaging regime of a single window), alongside the classical star-ratio statistic (Section 3.2, Table 1) computed directly from that same tree’s internal-node ages. Detectability is then a rank question, not a curve-smoothness question: for each of $N=30$ independently simulated chromosomes, we record what fraction of that chromosome’s own trees score as more selected-looking than the tree actually overlapping the true sweep site (rank 1 = most selected-looking tree on that chromosome; a top fraction near 0.5 is chance, near 0 is strong localization). All other settings ($n=10$, $N_e=1000$, chromosome length, midpoint sweep position, conversion of branch lengths to mutation units) match the first design.

4 Results

307

4.1 Sanity check: the neutral limit

308

We first check that the selection-augmented pipeline reduces correctly to the neutral coalescent when selection is absent, since an error here would undermine we trained and evaluated with s forced to exactly 0 (10 loci, $n=10$; 400 train / 80 validation / 80 prediction examples, $w=0.1$), where the expectation is simply the neutral coalescent: θ should be recoverable, and predicted s should be near 0 even though the neural network was never told $s=0$ was guaranteed. θ recovery reached $R^2=0.946$ (bias -7.9×10^{-4} , RMSE (root-mean-square error) 2.8×10^{-3}) across the prior range $\theta \in [0.001, 0.05]$, consistent with a 225-dimensional feature vector for $n=10$ samples encoding the ages and a vectorized Newick equivalent carrying ample information about θ once selection is not confounding. Predicted s correctly approaches 0 (mean 0.010, SD 0.016, against a true value

of exactly 0), confirming the selection-augmented pipeline reduces to the neutral case when $s=0$.

4.2 MSE vs. PGINN at 10 loci: effect of sample size

Every example pools 10 independently simulated loci into one candidate-region feature summary (Section 3.4). Figure 1 shows recovery for the first replicate at $n=10$ individuals ($n=20$: Figure 7, Appendix B, the same pattern at tighter scatter); Table 2 gives the 10-replicate mean $\pm$ SD with a paired t -test for the MSE vs PGINN treatment.

Selection strength is reported on the $\log(1+s)$ scale, on which the prior is uniform and the network is trained (Section 3.3). At $n=10$, PGINN improves s recovery in every one of the 10 replicates (R^2 0.806 $\rightarrow$ 0.861, RMSE 0.79 $\rightarrow$ 0.67, paired $p=8\times 10^{-5}$) and removes most of MSE’s downward bias ($-0.31 \rightarrow -0.13$). The price is a small loss in θ recovery (R^2 0.925 vs. 0.908, $p=0.012$). At $n=20$ the pattern is the same: PGINN is better for s in all 10 replicates (R^2 0.842 $\rightarrow$ 0.872, $p=6.5\times 10^{-5}$), while the θ difference is not significant ($p=0.16$). Doubling n from 10 to 20 improves both parameters for both losses without changing which loss wins for which parameter.

Table 2: 10 loci/example, independently simulated prediction set ($n=400$), mean $\pm$ SD across 10 replicates (seeds $\{67, \dots, 76\}$), at $n=10$ and $n=20$ sampled individuals, plus a paired t -test p -value on the 10 replicate-level PGINN –MSE differences (bold: $p < 0.05$).

n	Parameter	Loss	Bias (mean $\pm$ SD)	RMSE (mean $\pm$ SD)	R^2 (mean $\pm$ SD)	paired p
10	Θ	MSE	-0.0004 ± 0.0004	0.0033 ± 0.0002	0.925 ± 0.011	0.012
10	Θ	PGINN	-0.0011 ± 0.0003	0.0037 ± 0.0003	0.908 ± 0.015	
10	$\log(1+s)$	MSE	-0.31 ± 0.05	0.79 ± 0.02	0.806 ± 0.012	8.1e-05
10	$\log(1+s)$	PGINN	-0.13 ± 0.06	0.67 ± 0.05	0.861 ± 0.021	
20	Θ	MSE	-0.0003 ± 0.0004	0.0026 ± 0.0001	0.959 ± 0.004	0.158
20	Θ	PGINN	-0.0006 ± 0.0006	0.0030 ± 0.0010	0.938 ± 0.044	
20	$\log(1+s)$	MSE	-0.27 ± 0.08	0.76 ± 0.03	0.842 ± 0.012	6.5e-05
20	$\log(1+s)$	PGINN	-0.12 ± 0.07	0.68 ± 0.04	0.872 ± 0.015	

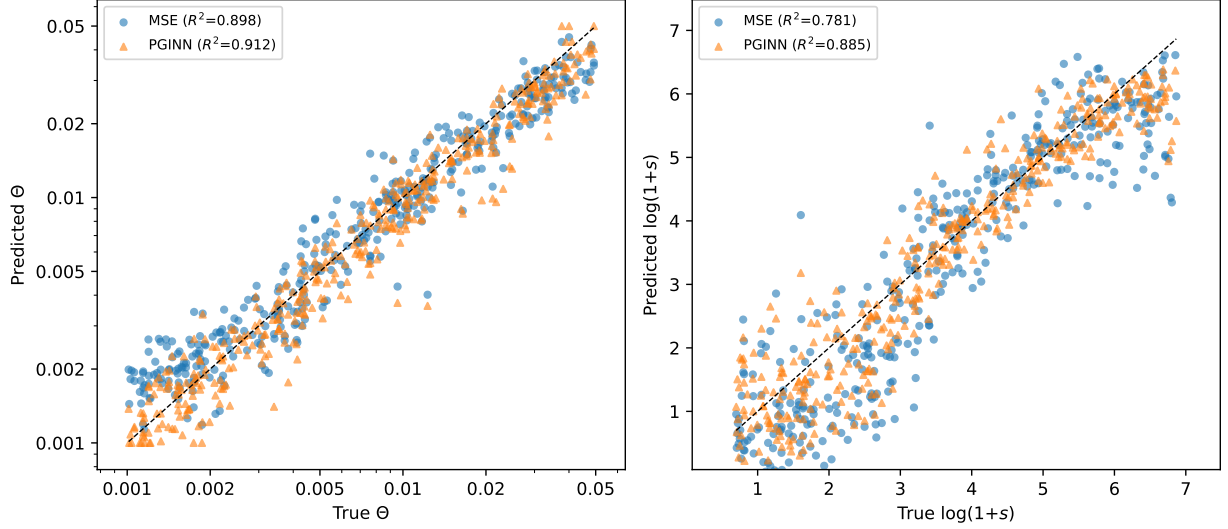


Figure 1: Predicted vs. true Θ (left) and s (right), 10 loci, $n=10$, replicate 1, 400-example independently simulated prediction set. Θ is shown on log axes, s on the $\log(1+s)$ scale. The PGINN fit (orange) tracks the diagonal more closely for s than plain MSE (blue), which underestimates strong selection; Θ recovery is similar for the two. See Table 2 for the 10-replicate mean $\pm$ SD summary this single replicate is drawn from.

4.3 MSE vs. PGINN at 1 locus: effect of sample size

332

At 10 loci the KL term's ages (u_k , Eq. 6) are an average over 10 realizations; at 1 locus they are the actual coalescence times of one genealogy. Figure 2 (Figure 8 in Appendix B for $n=20$) and Table 3 compare MSE and PGINN with a single genealogy per example, 10 replicates each. Both parameters are estimated less precisely than at 10 loci, θ most of all (R^2 about 0.70 at $n=10$ and 0.80 at $n=20$ for MSE, against 0.93 and 0.96 at 10 loci): θ sets the branch-length scale, which one genealogy estimates with large variance. The trade-off seen at 10 loci is sharper here. For s , PGINN is better in every replicate at both sample sizes (R^2 0.589 $\rightarrow$ 0.691 at $n=10$, $p=3.2 \times 10^{-5}$; 0.644 $\rightarrow$ 0.729 at $n=20$, $p=1.3 \times 10^{-4}$), with a smaller downward bias. For θ , MSE is better (0.702 vs. 0.674, $p=0.015$; 0.800 vs. 0.754, $p=0.00094$). The KL term weights each coalescent interval by how informative it is about the sweep-rescaled rate, which helps the estimate of the selection strength s at some cost to the estimate of the population-scaled mutation rate θ .

333
334
335
336
337
338
339
340
341
342
343

Table 3: 1 locus/example, independently simulated prediction set ($n=400$), mean $\pm$ SD across 10 replicates (seeds $\{67, \dots, 76\}$), at $n=10$ and $n=20$ sampled individuals, plus a paired t -test p -value on the 10 replicate-level PGINN –MSE differences (bold: $p < 0.05$, and the better R^2).

n	Parameter	Loss	Bias (mean $\pm$ SD)	RMSE (mean $\pm$ SD)	R^2 (mean $\pm$ SD)	paired p
10	Θ	MSE	-0.0009 ± 0.0003	0.0072 ± 0.0003	0.702 ± 0.023	0.015
10	Θ	PGINN	-0.0020 ± 0.0005	0.0075 ± 0.0003	0.674 ± 0.024	
10	$\log(1+s)$	MSE	-0.32 ± 0.09	1.22 ± 0.04	0.589 ± 0.024	3.2e-05
10	$\log(1+s)$	PGINN	-0.08 ± 0.05	1.06 ± 0.06	0.691 ± 0.034	
20	Θ	MSE	-0.0014 ± 0.0005	0.0060 ± 0.0004	0.800 ± 0.030	0.00094
20	Θ	PGINN	-0.0025 ± 0.0004	0.0066 ± 0.0002	0.754 ± 0.016	
20	$\log(1+s)$	MSE	-0.45 ± 0.10	1.13 ± 0.04	0.644 ± 0.028	0.00013
20	$\log(1+s)$	PGINN	-0.25 ± 0.05	0.98 ± 0.05	0.729 ± 0.029	

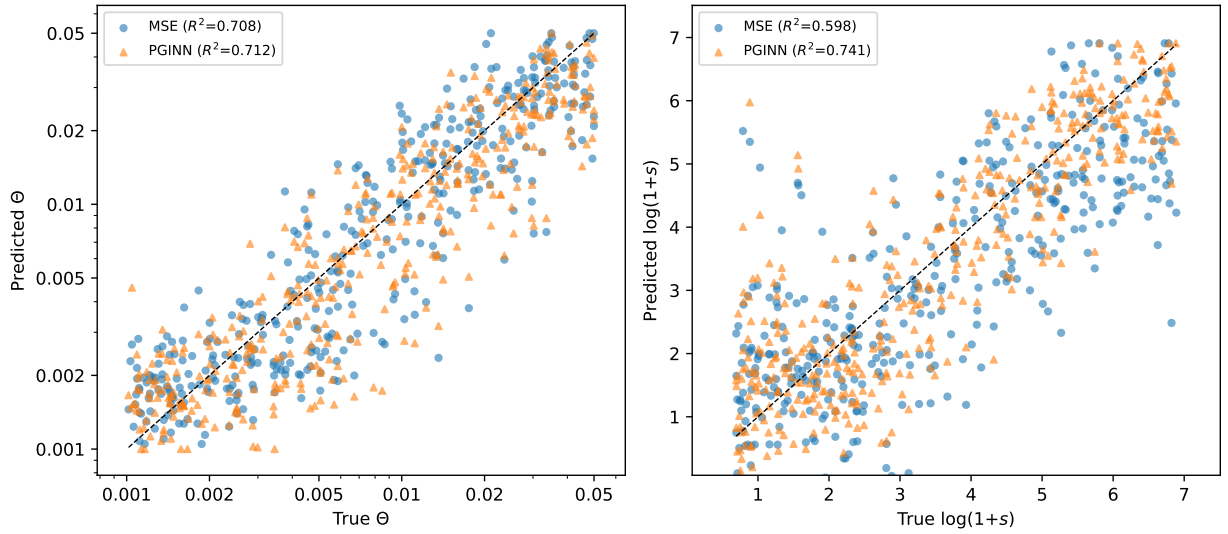


Figure 2: Predicted vs. true Θ (left) and s (right), 1 locus, $n=10$, replicate 1, 400-example independently simulated prediction set. Θ recovery is noisier than in the 10-loci regime for both losses; for s (on the $\log(1+s)$ scale) PGINN tracks the diagonal more closely than MSE. See Table 3 for the 10-replicate mean $\pm$ SD summary.

4.4 Detecting selection: neutral vs. selected loci

We repurpose the already-trained regression models as selection detectors: for each regime, we combine the selected prediction set (true s log-uniform on $[1, 1000]$) with a matched neutral one of the same size (true $s=0$, same θ prior and loci-per-example count), score both with the model’s predicted $\hat{s}$, and treat $\hat{s}$ as a detection statistic across all 10 replicates per regime/loss.

As a non-network baseline, we scored the same sets with the star-ratio already used for calibration (Section 3, Table 1), computed on the same ages the neural network sees: a single scalar summarizing how star-like the genealogy is, with no training and so no variation across replicates.

We summarize detection by the area under the receiver operating characteristic (ROC) curve,

the AUC: the probability that a randomly chosen selected example scores higher than a randomly
chosen neutral one (0.5 is chance, 1 is perfect separation). Table 5 shows the resulting AUCs and
per-bin sensitivities for all four regimes and all three detectors (ROC curves in Appendix C). No
detector exceeds an AUC of 0.87, because about a third of the selected set has $s < 10$, a range in
which sweep genealogies are hardly distinguishable from neutral ones (Table 1). At 10 loci the star
ratio is the best single detector (AUC 0.861 and 0.848 at $n=10$ and 20, against 0.81–0.83 for either
network). At 1 locus, the three are within 0.02 of each other (Table 5). Detection AUC does not
separate the two losses: the paired difference across replicates is not significant in three regimes,
and at 10 loci, $n=20$ MSE is slightly ahead (0.817 vs. 0.808, $p=0.002$).

Table 5 also reports sensitivity at a fixed 5% false-positive rate (the threshold is the 95th
percentile of each detector’s score on the neutral set), broken down by true selection strength.
Below $s=10$ every detector flags selected loci at about the false-positive rate, i.e. weak sweeps are
not detectable from genealogy shape. For $10 \leq s < 50$ the star ratio is the most sensitive detector at
10 loci (0.87 and 0.94, against 0.63–0.71 for the networks), and all three detect only 20–30% at 1
locus. For strong selection, the networks’ advantage appears where it matters most, with a single
genealogy: at 1 locus they detect 79–89% of loci with $50 \leq s < 200$ and 98–100% with $s \geq 200$, while
the star ratio detects only 52–60% of either.

To test whether a classical statistic makes a trained network unnecessary for detection, we
combined the star ratio and a network’s $\hat{s}$ in a logistic regression, evaluated by 5-fold cross-validation
(Table 4). At 10 loci, the combination adds nothing to the star ratio alone (AUC 0.860 for both at
 $n=10$; 0.847 vs. 0.848 at $n=20$; all AUCs in this paragraph are cross-validated, so the star ratios
differ from Table 5 in the third decimal). At 1 locus it helps in every replicate: star ratio + PGINN
reaches AUC 0.777 at $n=10$ and 0.812 at $n=20$ (star ratio alone 0.756 and 0.780), and it beats
PGINN alone by 0.03–0.04 ($p < 10^{-4}$ in both regimes). The choice of loss matters little here: star
ratio + PGINN exceeds star ratio + MSE by only 0.002 ($p=0.019$ at $n=10$, $p=0.09$ at $n=20$), and
adding MSE’s $\hat{s}$ as a third feature gives no further gain. So at 10 loci the star ratio alone suffices,
but with a single genealogy, a network’s $\hat{s}$ adds detection power that the star ratio does not have,
mostly for strong sweeps.

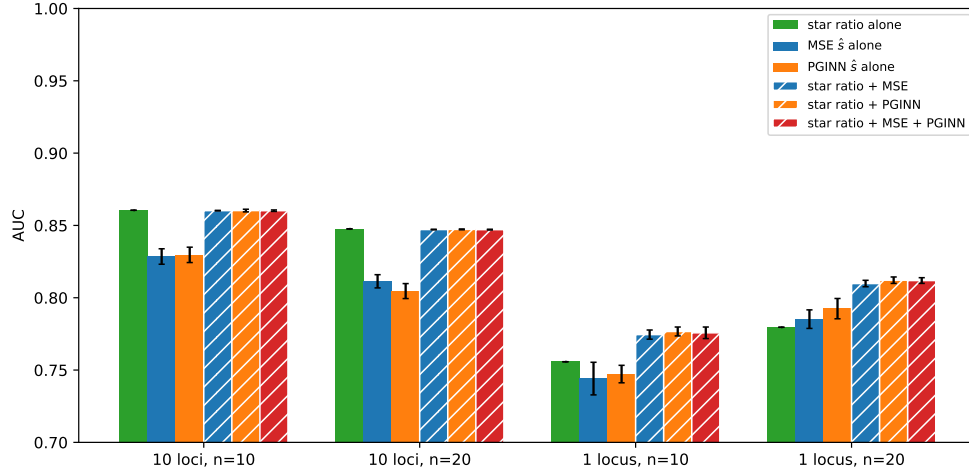


Figure 3: Detection AUC (5-fold cross-validated logistic regression; mean $\pm$ SD across 10 replicates) for each statistic alone (solid bars: star ratio, MSE $\hat{s}$, PGINN $\hat{s}$) and for the star ratio combined with one or both networks' $\hat{s}$ (hatched), across all four regimes. A network alone does not beat the star ratio: it is worse at 10 loci and about equal at 1 locus. Combining adds nothing at 10 loci but gives a reproducible gain of 0.02–0.03 in the single-locus regime. Which network's $\hat{s}$ is added matters little (star ratio + PGINN is ahead of star ratio + MSE by 0.002), and adding both gives nothing further.

Table 4: Detection performance of the star ratio alone and combined with network estimates of s . Each cell gives AUC/PR-AUC of a 5-fold cross-validated logistic regression, where PR-AUC is the area under the precision–recall curve (the average fraction of flagged examples that are truly selected, taken over all detection thresholds; chance level is the fraction of selected examples, here 0.5). Rows are regimes (number of loci / sample size n). Columns: *Star ratio*, the star ratio as the only predictor; + *MSE* $\hat{s}$ and + *PGINN* $\hat{s}$, the star ratio plus the $\hat{s}$ of the MSE-trained or the PGINN network; + *MSE* + *PGINN* $\hat{s}$, the star ratio plus both $\hat{s}$. Combined columns report mean $\pm$ SD across the 10 network replicates; the star-ratio column has a single value because it does not depend on a network. Bold marks the highest (mean) AUC in each regime, star ratio included (ties at three decimals all bold).

Loci/ n	Star ratio	+ MSE $\hat{s}$	+ PGINN $\hat{s}$	+ MSE + PGINN $\hat{s}$
10/10	0.860/0.901	0.860 $\pm$ 0.000/ 0.901 $\pm$ 0.000	0.860 $\pm$ 0.001/ 0.900 $\pm$ 0.002	0.860 $\pm$ 0.001/ 0.901 $\pm$ 0.000
10/20	0.848/0.894	0.847 $\pm$ 0.000/0.893 $\pm$ 0.000	0.847 $\pm$ 0.000/0.893 $\pm$ 0.000	0.847 $\pm$ 0.000/0.893 $\pm$ 0.000
1/10	0.756/0.754	0.775 $\pm$ 0.003/0.815 $\pm$ 0.004	0.777 $\pm$ 0.003/ 0.818 $\pm$ 0.009	0.776 $\pm$ 0.004/0.818 $\pm$ 0.008
1/20	0.780/0.764	0.810 $\pm$ 0.002/0.854 $\pm$ 0.003	0.812 $\pm$ 0.002/ 0.854 $\pm$ 0.005	0.812 $\pm$ 0.002/ 0.855 $\pm$ 0.003

4.5 Agreement between independently trained networks

Beyond average accuracy, a practical question is how much two independently trained networks disagree about the same genealogy, and whether that disagreement is informative: a genealogy whose shape is close to what a neutral locus would also produce is intrinsically hard to call, and a network's uncertainty should concentrate there. We treat each loss's 10-replicate ensemble (Sections 4.2–4.3) as an uncertainty proxy: for each example, the standard deviation of $\log(1+\hat{s})$ across the 10 replicates measures how much independently trained networks disagree on it, and we correlate this per-example replicate disagreement with the star ratio (Section 3) across all 400 selected examples in each regime.

Table 5: Detection AUC (mean $\pm$ SD across 10 replicates; classical baseline has no model/replicate dependence) and per-bin sensitivity: for each true- s bin, the fraction of truly-selected examples flagged at a 5% false-positive rate on the neutral set (networks: mean across the 10 replicates; subscript = number of selected examples in that bin). Each regime has 400 selected and 400 neutral examples. Bold marks the best AUC per regime (ties at three decimals all bold).

Loci	n	Detector	AUC (mean $\pm$ SD)	Sensitivity at 5% false positives, by true s , mean _(n)			
				[1, 10)	[10, 50)	[50, 200)	[200, 1000]
10	10	Classical: star ratio	0.861	0.05 ₍₁₂₄₎	0.87 ₍₁₁₀₎	1.00 ₍₇₇₎	1.00 ₍₈₉₎
10	10	MSE	0.831 $\pm$ 0.006	0.07 ₍₁₂₄₎	0.67 ₍₁₁₀₎	1.00 ₍₇₇₎	1.00 ₍₈₉₎
10	10	PGINN	0.832 $\pm$ 0.004	0.09 ₍₁₂₄₎	0.70 ₍₁₁₀₎	1.00 ₍₇₇₎	1.00 ₍₈₉₎
10	20	Classical: star ratio	0.848	0.13 ₍₁₄₅₎	0.94 ₍₈₉₎	1.00 ₍₇₆₎	1.00 ₍₉₀₎
10	20	MSE	0.817 $\pm$ 0.005	0.06 ₍₁₄₅₎	0.71 ₍₈₉₎	1.00 ₍₇₆₎	1.00 ₍₉₀₎
10	20	PGINN	0.808 $\pm$ 0.005	0.05 ₍₁₄₅₎	0.63 ₍₈₉₎	1.00 ₍₇₆₎	1.00 ₍₉₀₎
1	10	Classical: star ratio	0.757	0.07 ₍₁₅₂₎	0.23 ₍₇₉₎	0.52 ₍₇₅₎	0.52 ₍₉₄₎
1	10	MSE	0.753 $\pm$ 0.012	0.06 ₍₁₅₂₎	0.22 ₍₇₉₎	0.79 ₍₇₅₎	0.98 ₍₉₄₎
1	10	PGINN	0.757 $\pm$ 0.006	0.04 ₍₁₅₂₎	0.22 ₍₇₉₎	0.83 ₍₇₅₎	1.00 ₍₉₄₎
1	20	Classical: star ratio	0.780	0.07 ₍₁₂₈₎	0.20 ₍₉₀₎	0.53 ₍₇₇₎	0.60 ₍₁₀₅₎
1	20	MSE	0.796 $\pm$ 0.005	0.06 ₍₁₂₈₎	0.30 ₍₉₀₎	0.87 ₍₇₇₎	1.00 ₍₁₀₅₎
1	20	PGINN	0.801 $\pm$ 0.008	0.08 ₍₁₂₈₎	0.30 ₍₉₀₎	0.89 ₍₇₇₎	1.00 ₍₁₀₅₎

PGINN’s mean replicate disagreement is lower than MSE’s in every regime (12–22% reduction, Table 6). The two losses also differ in where the disagreement falls. For PGINN it is positively correlated with the star ratio in every regime ($r=0.29$ – 0.46): the networks disagree most on the less star-like, more neutral-looking genealogies, where s is weakly identified. For MSE the correlation is essentially zero (r between -0.14 and 0.05), so its disagreement is spread over easy and hard genealogies alike. Dropping each regime’s single worst MSE replicate changes MSE’s mean disagreement by at most 3% (Table 6, final column), so the difference is not caused by one poorly trained network.

Table 6: Mean per-example replicate disagreement (standard deviation of $\log(1+\hat{s})$ across the 10 replicates, on the $\log(1+s)$ scale; independently simulated selected prediction set, $n=400$ per regime), the PGINN-vs-MSE reduction, and each loss’s correlation between replicate disagreement and the star ratio. The final column recomputes MSE’s mean replicate disagreement after dropping its single worst- R^2 replicate for that regime.

Regime	MSE $\bar{\sigma}$	PGINN $\bar{\sigma}$	Reduction	corr(σ , star ratio)	
				MSE / PGINN	MSE $\bar{\sigma}$ (worst replicate out)
10 loci, $n=10$	0.27	0.23	16%	-0.10 / 0.39	0.26 (–3%)
10 loci, $n=20$	0.26	0.23	12%	0.02 / 0.46	0.25 (–2%)
1 locus, $n=10$	0.37	0.29	22%	0.05 / 0.33	0.36 (–3%)
1 locus, $n=20$	0.35	0.31	12%	-0.14 / 0.29	0.35 (–2%)

4.6 External validation: a hitchhiking scan simulated in msprime

Every result so far comes from our own coalescent simulator: single, unlinked loci generated under the deterministic hard-sweep rescaling of Section 2.2. A stronger test is to score the trained network on a different simulator entirely, with a genuinely different (stochastic, recombination-aware) sweep

model it never trained on.

We used msprime’s **SweepGenicSelection** model (Baumdicker et al., 2022), which implements the same hard-sweep coalescent approximation (Braverman et al., 1995) that motivates our own model (Section 2.2), but with recombination breaking down linkage around the selected site instead of treating every locus as independent. We simulated a chromosome with a hard sweep fixed at its midpoint ($n=10$ samples, matching a trained model), sliced it into 100 windows, and averaged each window’s feature vector over 30 independent replicate simulations, exactly as for the 10-loci training data. Each window’s local tree was exported to Newick and converted to our feature format with the same code used to score any externally-estimated tree. Branch lengths were converted from generations to the branch-length scale of Eq. 5 as described in Section 3.6.

Figure 4 shows the result for two recombination rates on the same 500 kb chromosome ($r=2\times 10^{-8}/\text{bp}$ and $r=1.2\times 10^{-8}/\text{bp}$, both with $s_g=0.01$). msprime’s sweep model differs from ours in its trajectory and in its treatment of recombination, so we do not expect an exact numerical correspondence between its s_g and the network’s s ; the comparison is about shape. Both losses recover the hitchhiking signature at both rates, predicted s rising toward the true sweep site and falling off with distance in both directions, from a far-field level of about 1–2 to a peak of about 30 (PGINN) or 40–50 (MSE). Consistent with hitchhiking, the lower recombination rate gives a broader peak and the higher rate a sharper, more localized one. A network trained exclusively on our own non-recombining, deterministic sweep model still recognizes the genealogical footprint of a recombination-linked hard sweep it was never exposed to, and responds to a change in recombination rate.

We back this with three quantitative checks, each replicated across 3 independent msprime simulations (different random seeds) rather than reported from a single run. First, predicted s is negatively correlated with distance from the true sweep site in every one of the 12 (simulation $\times$ recombination rate $\times$ loss) combinations (Pearson’s $r = -0.55$ to -0.73 , somewhat stronger for PGINN than for MSE in every combination). We report the correlation and its replication as the evidence, not the accompanying p -values ($p < 10^{-8}$ throughout), since the 100 windows within one replicate are adjacent positions on a recombining chromosome, not independent draws. Second, a peak half-width statistic (the contiguous span over which predicted s stays above the midpoint between its on-sweep peak and its far-field baseline) shows the broadening: for PGINN the footprint is 25 kb wide at $r=2\times 10^{-8}/\text{bp}$ in all 3 simulations and 45–50 kb at $r=1.2\times 10^{-8}/\text{bp}$; MSE’s half-width varies more between simulations (30–45 kb and 40–50 kb). Third, a no-sweep neutral control at $r=2\times 10^{-8}/\text{bp}$ (same chromosome, neutral coalescent, same 3 seeds) checks whether the network manufactures a spurious peak from neutral variation. It does not: predicted s stays at the level of the sweep panels’ far field for both losses (mean 1.3–1.5, maximum 2.9–3.4 across simulations and losses), an order of magnitude below the on-sweep peaks (28–50).

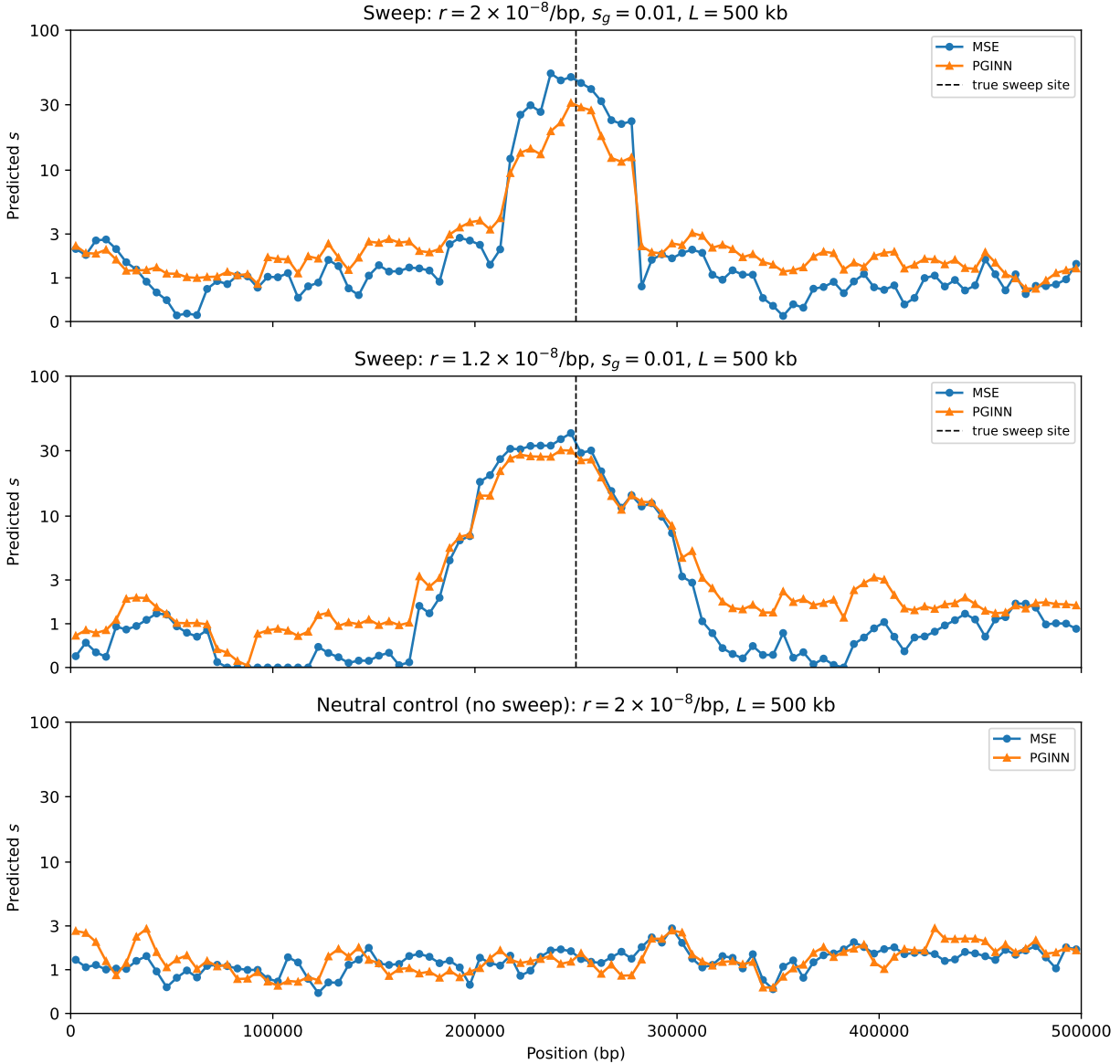


Figure 4: Predicted s vs. genomic position for a 500 kb chromosome simulated in msprime, averaged over 30 independent replicate simulations per window; all three panels share the same x - and y -axis ranges, with s on a $\log(1+s)$ -spaced axis labeled in s units. Top: recombination rate $r=2\times 10^{-8}/\text{bp}$, hard sweep at the midpoint (dashed line; msprime’s per-generation selection coefficient $s_g=0.01$). Middle: $r=1.2\times 10^{-8}/\text{bp}$, same sweep. Bottom: Neutral control, same chromosome and recombination rate as the top panel but no sweep.

4.7 A harder test: detection from a single, unaveraged chromosome

437

The result above averages 30 independent replicate chromosomes per window; a genome scan has only one. A single 500 kb chromosome at this recombination rate and sample size contains few distinct local trees (median 34), far fewer than the 100 window positions scored above, so a finer grid only repeats the same trees’ scores (73% of 5 kb windows fall entirely on one tree). Changing

438

439

440

441

the window width or the recombination rate does not help: higher rates give more trees per window but narrow the sweep footprint in step, so the per-window sampling noise (Appendix E) stays about the same.

Instead, for each of 30 single-chromosome replicates, we rank all local trees by each statistic and record the rank of the tree at the true sweep site as a fraction of the chromosome’s trees (0 = best-ranked, 0.5 = chance; Section 3.6). All three statistics place the sweep tree better than chance, but none localizes it reliably (Table 7, Figure 5): its median rank is in the top 15.3% for the star ratio, 20.4% for PGINN and 24.5% for MSE, and it is among the top 20% of trees on 57%, 50% and 40% of chromosomes. Appendix F shows three example chromosomes.

The star ratio, which uses only the topmost split, ranks the sweep tree higher than MSE (paired Wilcoxon test over the 30 chromosomes, $p=0.008$); its advantage over PGINN ($p=0.20$) and PGINN’s over MSE ($p=0.17$) are not significant with 30 chromosomes. Unlike on our own simulator (Section 4.4), combining the star ratio with a network’s $\hat{s}$ does not help: the logistic classifiers fitted on our 1-locus prediction set, applied unchanged to the msprime trees, place the sweep tree at a median of 20.3% (PGINN) and 20.4% (MSE), no better than the networks alone; the star ratio alone ranks the sweep tree higher than star ratio + MSE ($p=0.011$), but not significantly higher than star ratio + PGINN ($p=0.16$). On msprime’s recombining sweeps, only the MSE-based detectors fall significantly behind the star ratio; PGINN, alone or combined with the star ratio, is statistically indistinguishable from it.

Table 7: Rank of the true sweep site’s own local tree among all trees scored on that chromosome, across 30 independent single-chromosome replicates (median 34 trees/chromosome), for each statistic alone and for the star ratio combined with a network’s $\hat{s}$ by the logistic classifiers of Section 4.4, fitted on our own 1-locus, $n=10$ prediction set and applied unchanged. Top-% = $100 \times \text{rank} / (\text{trees on that chromosome})$; 0% is the single best-ranked tree, 50% is chance.

Detector	median top-%	mean top-%	top 10%	top 20%	top 25%
Star ratio (classical)	15.3	21.3	30.0	56.7	66.7
MSE predicted s	24.5	31.8	20.0	40.0	53.3
PGINN predicted s	20.4	25.9	30.0	50.0	63.3
Star ratio + MSE $\hat{s}$	20.4	25.5	20.0	50.0	63.3
Star ratio + PGINN $\hat{s}$	20.3	23.9	26.7	50.0	66.7

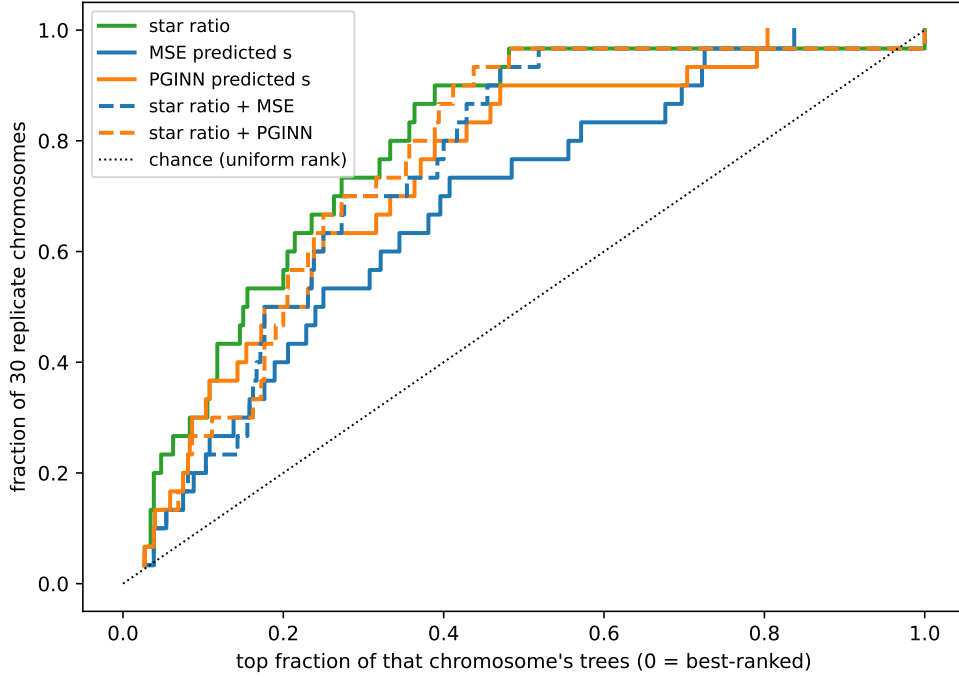


Figure 5: Cumulative distribution, across the 30 single-chromosome replicates of Table 7, of the rank of the true sweep site’s tree among all trees on its chromosome, as a fraction of that chromosome’s trees (0= best-ranked). A point (x, y) means that on a fraction y of the chromosomes the sweep tree ranks within the top x ; curves further up and to the left are better, and the dotted diagonal is chance. Solid lines: each statistic alone (green: star ratio; blue: MSE $\hat{s}$; orange: PGINN $\hat{s}$). Dashed lines: the star ratio combined with the MSE (blue) or PGINN (orange) $\hat{s}$. The value at $x=0.2$ is the “top 20%” column of Table 7.

4.8 External validation: *Drosophila melanogaster* ADH locus

Section 4.6 tested generalization to a different simulator with a known, if different, generative model. Real biological data, where the genealogy itself must be estimated, is an additional challenge. We scored the alcohol dehydrogenase (*Adh*) region, one of the early genes for which within-species nucleotide polymorphism was surveyed in a landmark study that sequenced 11 alleles and found only 1 of 43 polymorphisms changed an amino acid, the Fast/Slow electrophoretic variant (Kreitman, 1983). We used a much larger modern sample, 176 lines from a single wild-caught *D. melanogaster* population (Drosophila Genetic Reference Panel, DGRP; Mackay et al., 2012), the same species and locus but a different population from Kreitman’s original five, pooled populations. We used the sequences of DGRP Freeze 2.0 (Huang et al., 2014). Of its 205 lines, 29 have no sequence data in this region (every position missing) and were excluded, leaving 176; in these, a median of 1.4% of positions (at most 12.8%) are missing.

We used MIGRATE, a Bayesian Markov chain Monte Carlo sampler (Beerli et al., 2022), which integrates over random coalescent genealogies to estimate $\Theta=4N\mu$ from sequence data, to sample genealogies per locus as the input data for PGINN; migrate’s own per-locus Θ estimates gave an independent, non-neural network check throughout. A four-gamete test (Hudson and Kaplan, 1985) on the alignment found pervasive recombination (Hudson-Kaplan $R_m=14$ minimum events among 108 segregating sites across 3351 bp), so the region was cut at these 14 breakpoints into 15 segments

(51–833 bp) with no four-gamete violation inside any segment, each cut placed midway between the two segregating sites that bound it. Each segment was treated as an independent locus in one joint MIGRATE run (version 6.1.11), with no pooling. Migrate’s own genealogy sampler assumes a neutral coalescent.

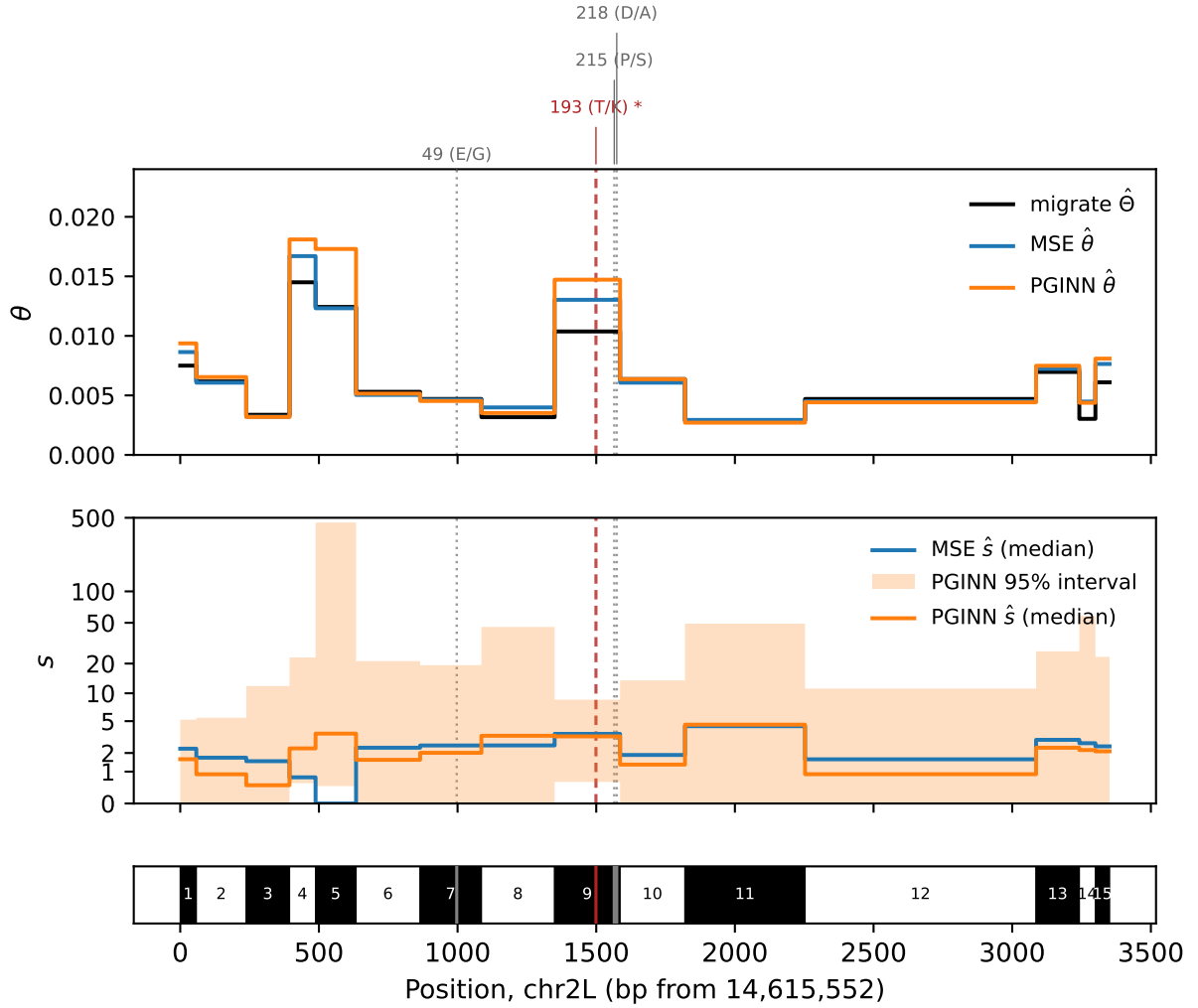
The network scores one genealogy at a time. Rather than a single genealogy per segment, we scored a sample from MIGRATE’s posterior distribution of genealogies: 10 replicate chains of 10^6 steps each, keeping every 100th recorded genealogy, for 1000 genealogies per segment, and report per segment the median estimate over these genealogies with the central 95% interval. This interval reflects uncertainty about the genealogy only, not the network’s own estimation error. MIGRATE’s genealogies are in expected mutations per site, the branch-length scale u of Eq. 5, and its Θ is the same population-scaled mutation rate as our θ ; we score the genealogies as they are, without rescaling, so the network’s $\hat{\theta}$ and MIGRATE’s Θ are directly comparable.

We scored the genealogies with networks trained at the full sample size ($n=176$, 1 locus per example, the priors of Section 3.2; 10 replicates each). On simulated data these recover θ well ($R^2=0.958 \pm 0.011$ for MSE, 0.944 ± 0.011 for PGINN) and s moderately ($\log(1+s)$ $R^2=0.72$ for both); unlike at $n=10$ and $n=20$, PGINN does not improve on MSE for s at this sample size (paired $p=0.97$), and MSE is slightly better for θ ($p=0.028$).

MIGRATE’s posterior genealogies are not fully neutral-like. For each segment we compared the $n-1$ coalescent-event ages of 20 random posterior genealogies (the quantity PGINN’s loss is built on, Section 2.4) with 50 neutral genealogies simulated at the same sample size and the segment’s MIGRATE Θ . In every segment, 2 to 13 of the 20 have at least one age more than 5 standard deviations from the neutral reference, mostly among the intermediate coalescent events, which the neutral coalescent constrains tightly at this sample size.

Table 8 and Figure 6 show all 15 segments. The network’s $\hat{\theta}$ agrees closely with MIGRATE’s independent Θ on the same absolute scale (Spearman $\rho=0.96$ for both losses; the medians differ by less than 20% at 11 of 15 segments for MSE and 9 of 15 for PGINN, and by at most 48%). None of the segments shows evidence of a hard sweep. The median $\hat{s}$ is between 0 and 5 at every segment for both losses, the range in which sweep genealogies are indistinguishable from neutral ones (Table 1), and at all but one segment 13 to 64% of the posterior genealogies give PGINN $\hat{s}<1$. Only segment 9 has a 95% interval that excludes 0, and it too lies below 10. The upper ends of the intervals reach about 60 (451 for PGINN at segment 5). $\hat{s}$ is unrelated to MIGRATE’s Θ across segments (Spearman $\rho=0.08$ for PGINN, -0.40 for MSE, neither significant).

Translating the coding exons (verified against the current *D. melanogaster* reference annotation) across all 176 lines finds 4 variable codons that change an amino acid, against 15 that do not, one of which, codon 193 (chr2L:14617050–14617052, Thr/Lys), is Kreitman’s original Fast/Slow polymorphism. Codon 49 (Glu/Gly) falls inside segment 7, and codon 193 and the remaining two nonsynonymous codons (215, Pro/Ser; 218, Asp/Ala) all fall inside segment 9. Segment 9 is the one segment whose $\hat{s}$ interval excludes 0 (PGINN median 3, interval 1–9), and it has the third-highest MIGRATE Θ of the 15 segments.



* Kreitman (1983) Fast/Slow site

Figure 6: Table 8's estimates along the *Adh* locus, x -axis in bp from the start of the sequenced chr2L region; each segment's estimate is drawn as a step across its physical span. Top: θ , migrate's independent estimate (black) against the MSE (blue) and PGINN (orange) medians, all on the same scale. Middle: s , MSE and PGINN medians over 1000 posterior genealogies per segment, with PGINN's 95% interval shaded, on a $\log(1+s)$ -spaced axis labeled in s units (MIGRATE assumes a neutral coalescent and has no s). Bottom: the black/white track marking the 15 segment boundaries (numbered as in Table 8). Vertical lines mark the 4 nonsynonymous codons found in the 176-line alignment (gray) with Kreitman's (1983) Fast/Slow site (codon 193) distinguished in red.

Table 8: The *Adh* locus (*D. melanogaster*, DGRP), all 176 sampled lines, the 15 segments cut at the four-gamete breakpoints (no pooling), scored with the $n=176$, 1-locus models (replicate 1), against MIGRATE’s independent per-segment $\hat{\Theta}$ (posterior mode). Network estimates are medians over 1000 posterior genealogies per segment, with the central 95% interval for $\hat{s}$ in parentheses; the last column gives the percentage of genealogies with PGINN $\hat{s} < 1$. Network $\hat{\theta}$ and MIGRATE’s Θ are on the same scale (Section 4.8).

Seg.	bp	migrate $\hat{\Theta}$	MSE $\hat{\theta}$	MSE $\hat{s}$	PGINN $\hat{\theta}$	PGINN $\hat{s}$	% $\hat{s} \approx 0$
1	58	0.0075	0.0086	2 (0–8)	0.0094	2 (0–5)	32
2	180	0.0062	0.0061	2 (0–7)	0.0065	1 (0–5)	54
3	156	0.0034	0.0032	2 (0–17)	0.0032	0 (0–12)	64
4	94	0.0145	0.0167	1 (0–8)	0.0181	2 (1–23)	13
5	146	0.0124	0.0123	0 (0–7)	0.0173	4 (0–451)	14
6	231	0.0053	0.0050	2 (0–22)	0.0052	2 (0–21)	39
7	221	0.0047	0.0046	3 (0–17)	0.0045	2 (0–19)	34
8	264	0.0032	0.0040	3 (0–32)	0.0035	3 (0–46)	23
9	236	0.0104	0.0130	4 (1–8)	0.0147	3 (1–9)	5
10	234	0.0064	0.0061	2 (0–14)	0.0064	1 (0–14)	42
11	433	0.0029	0.0029	4 (0–53)	0.0027	5 (0–49)	17
12	833	0.0047	0.0045	2 (0–8)	0.0044	1 (0–11)	53
13	156	0.0070	0.0072	3 (0–24)	0.0075	2 (0–26)	22
14	58	0.0030	0.0045	3 (0–62)	0.0044	2 (0–58)	29
15	51	0.0061	0.0076	2 (0–19)	0.0081	2 (0–23)	29

5 Discussion

The theory term improves the estimate of the parameter it encodes. PGINN estimates the selection strength s more accurately than MSE in all four regimes and in every one of the 10 replicates (Sections 4.2–4.3), mostly by removing MSE’s underestimation of strong selection. The cost falls on θ : its R^2 is slightly lower with the theory term, significantly so in three of the four regimes, and its point estimate is shifted downward, most strongly at 1 locus (paired t -test on the per-replicate bias: 10 loci, $n=10$, $p=0.0015$; 1 locus, $p=0.00014$ and $p=0.00069$ at $n=10$ and 20; 10 loci, $n=20$, $p=0.11$). A plausible explanation is that the KL term compares the sweep-rescaled rate interval by interval, which rewards the shape of the genealogy (informative about s) more than its overall scale (informative about θ); at 1 locus the ages it is evaluated on are also a single noisy realization. Independently trained PGINN networks agree more with each other than MSE networks do, and they disagree mainly on neutral-looking genealogies, where s is weakly identified; MSE networks disagree on easy and hard genealogies alike (Section 4.5).

For detecting selection, PGINN and MSE are indistinguishable; the relevant comparison is with the star ratio (Section 3). At 10 loci the star ratio is as good as or better than either network. With a single genealogy the networks detect strong sweeps that the star ratio misses, and combining the star ratio with a network’s $\hat{s}$ beats the star ratio alone in every replicate. No detector finds weak sweeps ($s < 10$). On msprime’s recombining sweeps, however, the combination no longer improves on the star ratio (Section 4.7), so this gain depends on how closely the data match the training model.

Sampling more individuals per locus helps in every regime: from $n=10$ to $n=20$, θR^2 rises from 0.93 to 0.96 at 10 loci and from 0.70 to 0.80 at 1 locus (MSE), and s recovery improves for both losses. The gain is smaller than that from adding independent loci; more loci and more individuals

per locus are not interchangeable sources of information.

Several simplifications limit how far these results generalize. The model is a deterministic hard sweep at a single locus, funneled through one founding lineage (Section 2.3); a soft sweep from standing variation or recurrent mutation produces less star-like genealogies for the same s . The population-scaled $s=cs_g$ (Section 2.1) is not decomposed into population size and per-generation coefficient. The KL term evaluates the sweep-rescaled rate at the observed ages instead of integrating the full inhomogeneous-Poisson-process likelihood over each interval. These choices keep the question about the loss function rather than the simulator; soft sweeps or an ancestral-selection-graph treatment are natural extensions.

A genome scan also faces a limit independent of the network. The external validation in Figure 4 averages 30 replicate simulations per window, but a scan has one chromosome, which at this recombination rate contains only a few dozen distinct local trees; no window width or recombination rate recovers the averaged curve from a single chromosome (Section 4.7). Scored tree by tree, the tree at the true sweep site ranks at the median in the top 15 to 25% of its chromosome’s trees: detection from one chromosome is possible but weak, and only the MSE-based detectors fall significantly behind the star ratio.

Sequence data without a known truth are a harder test. For the 15 *Adh* segments (Section 4.8) we scored 1000 posterior genealogies per segment from MIGRATE’s Bayesian sampler, which carries genealogical uncertainty into each estimate. The networks’ $\hat{\theta}$ agrees with MIGRATE’s Θ , so they read the branch-length scale of genealogies they did not simulate. Some posterior genealogies in every segment, however, have intermediate coalescent events far from the neutral expectation, whether from features of the data (population structure, selection, or recombination within segments) or from MIGRATE’s sampler, so these estimates partly extrapolate beyond the training distribution.

No segment shows evidence of a hard sweep: the median $\hat{s}$ is between 0 and 5 everywhere, a range the calibration cannot distinguish from neutral, and the wide upper ends of the intervals reflect genealogical uncertainty in short or diverse segments. This is not a recovery result, since the true s is unknown, and it bears only weakly on the history of the *Adh* Fast/Slow polymorphism (Kreitman, 1983). Its excess of linked silent polymorphism was interpreted as evidence of balancing selection (Hudson et al., 1987; Kreitman and Hudson, 1991); later work questioned this, suggesting that the Fast allele is a derived variant that recently swept to intermediate frequency (Begun et al., 1999; Charlesworth and Charlesworth, 2017), a partial sweep that our model of a completed hard sweep does not include. The networks have little power for $s<10$ (Section 4.4), their training data contain no balancing selection, and an old sweep whose genealogy has relaxed toward neutral shape lies outside the model. Segment 9, which contains the Fast/Slow site, is the only segment whose $\hat{s}$ interval excludes 0, but its estimate is small and its diversity among the highest in the region, which does not fit a recent hard sweep centered there.

Applying McDonald and Kreitman’s (1991) test to our 176-line DGRP sample, with their *D. simulans*/*D. yakuba* comparison unchanged, removes their significant excess of replacement fixations (Fisher exact $p=1.0$ vs. their $p=0.007$; Appendix G). Several sites that appear fixed among their 11 sequences, including the Fast/Slow site, are polymorphic among 176. This is a caution about small-sample tests for selection, not evidence against selection at *Adh*: the divergence side of the test is unchanged, and our sample differs from theirs in sampling scheme as well as size.

For theory-informed losses more generally, the fixed-ratio mixture needs its weight w chosen on

validation data, and with care, since a validation-only sweep can favor a value that does not hold up across replicates (Appendix A). The MSE component is necessary: at $w=1$ (no MSE term), recovery of both parameters and detection collapse (Appendix A). An adaptive mixture, in which the balance between data and theory shifts with data informativeness or is learned, would remove the manual choice of w and is a natural next step.

Data and Code Availability

All code, a `README` with build and reproduction instructions, and a `requirements.txt` dependency list are available at <http://github.com/pbeerli/pginn>, released under the MIT license. The trained model weights and scalars needed to reproduce the paper’s figures and tables, and the MIGRATE posterior genealogies of the *Adh* segments, are deposited at Zenodo (<https://doi.org/10.5281/zenodo.23069672>); a script in the repository downloads them. Every figure and table in this paper can be reproduced from the code in that repository; `reproducibility.txt` there maps each one to the specific script and invocation that produced it. The larger generated simulation datasets are not deposited directly due to size but are fully and exactly regenerable from the seeded data-generation scripts and commands documented in the repository.

Acknowledgements

This work was supported by NSF award DBI-2549631. Claude Code (Anthropic, Claude Sonnet 5/Opus 5.5) assisted with code implementation, running experiments, and manuscript drafting under the author’s direction. The research idea, model design, and interpretation of the results are the author’s own.

A The Loss Weight w : Validation-Set Selection

The total training objective (Section 2.4) mixes the MSE and PGINN loss terms in a fixed ratio w . We chose w by a sweep over $w \in \{0.05, 0.1, 0.2, 0.3\}$ in all four regimes (replicate 1), scored on the 100-example validation split, never on the independently simulated prediction set the rest of the paper reports on, so the prediction-set comparisons in the main text are not biased by this hyperparameter search. Table 9 gives the full sweep. $w=0.1$ gives the best validation $\log(1+s) R^2$ in three of the four regimes; at 10 loci, $n=10$, $w=0.2$ is marginally better (0.872 vs. 0.861, a difference within single-replicate noise). We therefore use the single value $w=0.1$ throughout rather than tuning w per regime.

We also checked the extreme $w=1$ (pure PGINN, no MSE term, replicate 1 only, all four regimes): θ recovery drops sharply (R^2 0.26–0.66 against 0.71–0.95 at $w=0.1$), s recovery collapses below what predicting the mean would achieve ($\log(1+s) R^2$ between -1.8 and -0.8), and detection AUC falls to 0.50–0.68. The KL term alone does not identify the parameters; the mixture’s MSE component is doing necessary work, not merely diluting a purer PGINN signal.

Table 9: PGINN w sweep, replicate 1, selected on the 100-example validation split (not the independently simulated prediction set that the rest of this paper reports on, avoiding hyperparameter-selection leakage). Bold marks the best validation $\log(1+s)$ R^2 per regime.

Regime	w	θ R^2 (val)	$\log(1+s)$ R^2 (val)	$\log(1+s)$ RMSE (val)
10 loci, $n=10$	0.05	0.935	0.838	0.92
10 loci, $n=10$	0.10	0.912	0.861	0.85
10 loci, $n=10$	0.20	0.870	0.872	0.82
10 loci, $n=10$	0.30	0.886	0.842	0.91
10 loci, $n=20$	0.05	0.935	0.832	0.93
10 loci, $n=20$	0.10	0.926	0.849	0.88
10 loci, $n=20$	0.20	0.872	0.829	0.93
10 loci, $n=20$	0.30	0.876	0.815	0.97
1 locus, $n=10$	0.05	0.763	0.697	1.28
1 locus, $n=10$	0.10	0.765	0.712	1.25
1 locus, $n=10$	0.20	0.771	0.704	1.27
1 locus, $n=10$	0.30	0.744	0.684	1.31
1 locus, $n=20$	0.05	0.736	0.741	1.17
1 locus, $n=20$	0.10	0.743	0.743	1.16
1 locus, $n=20$	0.20	0.732	0.699	1.26
1 locus, $n=20$	0.30	0.728	0.707	1.24

B Recovery Scatter Plots at $n = 20$

618

The $n=20$ recovery scatter plots for both loci counts are collected here rather than in the main text: in both cases the pattern is the same as the corresponding $n=10$ figure (Figure 1, Figure 2) at a tighter overall scatter, and the quantitative comparison a reader actually needs is already in Table 2 and Table 3.

619

620

621

622

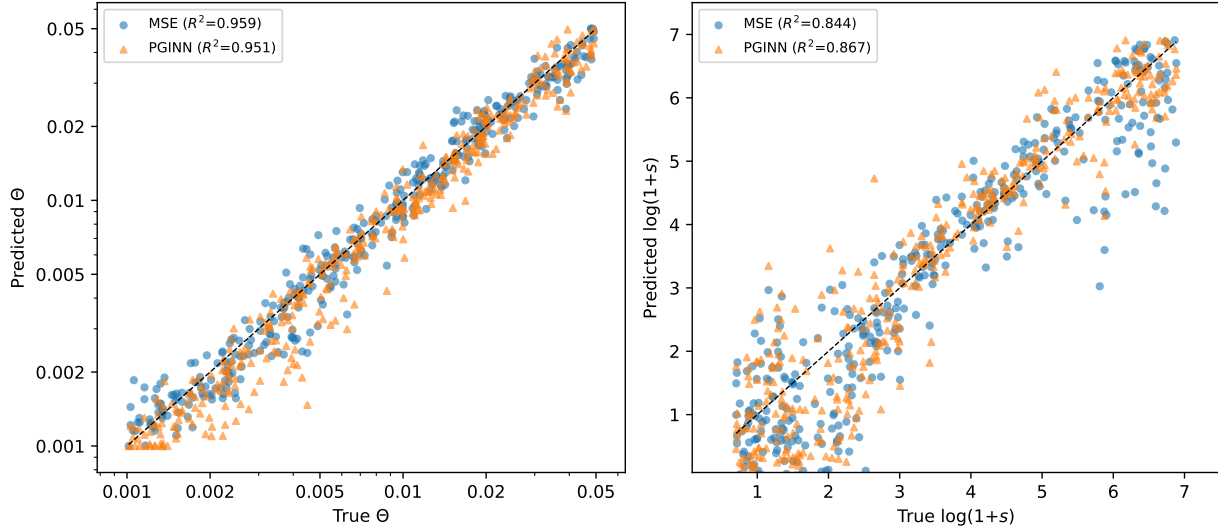


Figure 7: Predicted vs. true Θ (left) and s (right), 10 loci, $n=20$, replicate 1, 400-example independently simulated prediction set. The same pattern as Figure 1 at a tighter overall scatter for both losses; per Table 2, the s -recovery gap here is small but, with 10 replicates, statistically significant.

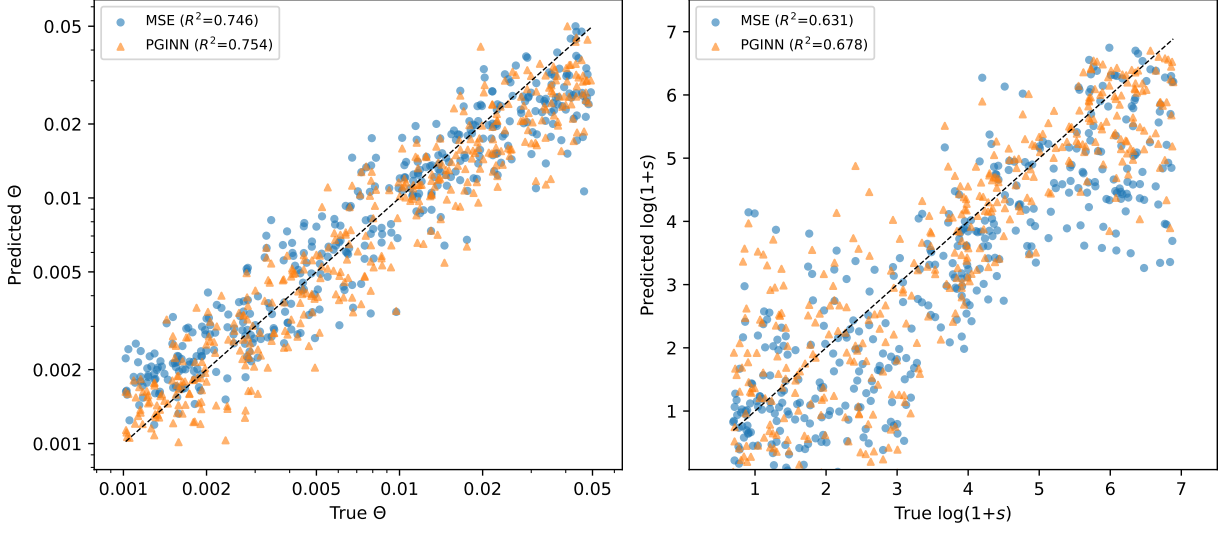


Figure 8: Predicted vs. true Θ (left) and s (right), 1 locus, $n=20$, replicate 1, 400-example independently simulated prediction set, with the same axes as Figure 2.

C ROC Curves for Selection Detection

Table 5 (Section 4.4) reports the AUC each detector's ROC curve summarizes; the curves themselves are collected here.

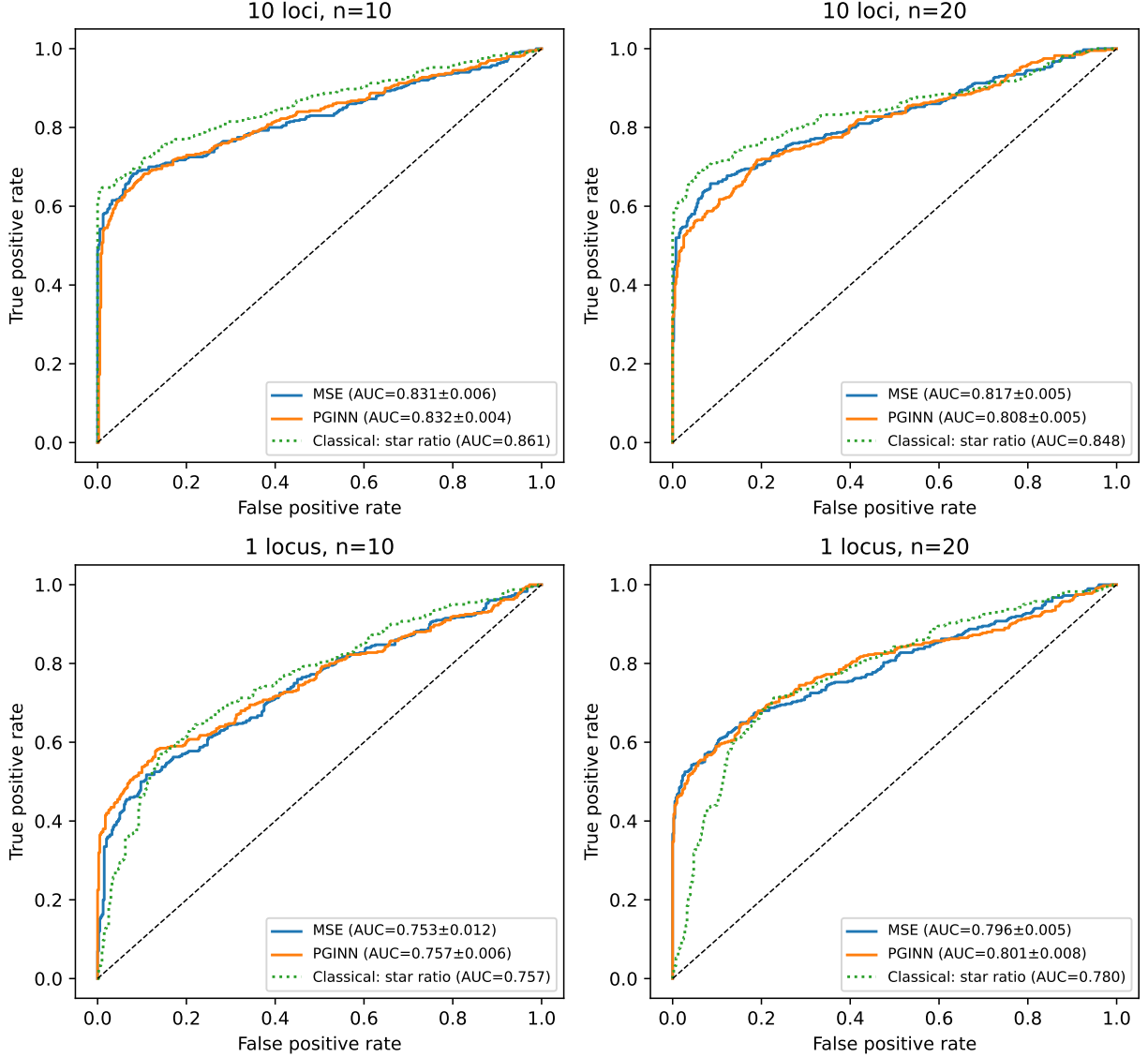


Figure 9: ROC curves (replicate 1) for detecting selection (true $s > 0$) vs. neutrality (true $s = 0$), using the regression models' predicted $\hat{s}$ (MSE and PGINN networks, legend AUC is the 10-replicate mean $\pm$ SD) and the classical star-ratio statistic as detection scores, across all four regimes (10 loci or 1 locus, $\times$ $n=10$ or $n=20$).

D Star Ratio: Definition and Sample-Size Dependence

Section 3 defines the star ratio as $(t_K - t_{K-1})/t_K$, where t_K is the TMRCA and t_{K-1} the second-deepest coalescence age (ages sorted descending, $K=n-1$). Under the standard neutral coalescent (Kingman, 1982), with T_k the independent waiting time at k lineages ($T_k \sim \text{Exp}(k(k-1)/2)$) and $t_K = \sum_{k=2}^n T_k$, the gap $t_K - t_{K-1}$ is exactly T_2 : the single interval during which the last two lineages coalesce. So the star ratio is T_2/t_K , a closed-form quantity whose expectation can be checked by simulation independently of the simulator used elsewhere in this paper. Because $E[T_2]=1$ regardless

of n , while $E[t_K]=2(n-1)/n$ converges to 2 as $n \rightarrow \infty$, the neutral expectation of this ratio does not approach 1 as n grows; large-trial simulation (500,000 draws from the T_k distributions directly, no coalescent simulator) gives $E[\text{star ratio}] \approx 0.470$ at $n=10$, 0.442 at $n=20$, and 0.423 at $n=100$. These values agree, within sampling error, with the $s=0$ rows of Table 1 (0.450 at $n=10$ and 0.440 at $n=20$, each from 200 genealogies, standard error about 0.017) and with the noisier $s=0$ rows of Table 10 below (10 genealogies per cell). Under a completed hard sweep, the whole genealogy instead compresses toward the sweep's origin (Kaplan et al., 1989), so $t_{K-1} \rightarrow t_K$ and the ratio $\rightarrow 0$; it is this convergence toward zero, not the specific neutral value above, that makes a small star ratio the signature of a star-like, swept genealogy.

Table 10 extends this check to $n \in \{10, 20, 100\}$ across the same s range as Table 1, but averaged over only 10 replicate genealogies per cell (`code/star_ratio_appendix.py`), instead of Table 1's 200. The qualitative Kingman-to-star interpolation still holds at every n , but the standard deviations are large relative to the gap between adjacent s values, and are not smaller at $n=100$ than at $n=10$: increasing the sample size per tree does not reduce the noise that comes from averaging over few replicate trees. That noise source is what Table 1's 200 replicates are chosen to control.

Table 10: Star ratio mean $\pm$ SD across 10 replicate genealogies per cell (`code/star_ratio_appendix.py`, $\theta=0.02$), at three sample sizes. Compare to Table 1's 200-replicate $n=10$ and $n=20$ rows.

n	s	star ratio (mean $\pm$ SD, 10 reps)
10	0	0.543 $\pm$ 0.175
	1	0.497 $\pm$ 0.281
	4	0.478 $\pm$ 0.192
	20	0.345 $\pm$ 0.200
	100	0.073 $\pm$ 0.059
	400	0.058 $\pm$ 0.032
	2,000	0.075 $\pm$ 0.045
20	0	0.315 $\pm$ 0.204
	1	0.381 $\pm$ 0.313
	4	0.676 $\pm$ 0.109
	20	0.187 $\pm$ 0.172
	100	0.090 $\pm$ 0.043
	400	0.066 $\pm$ 0.044
	2,000	0.038 $\pm$ 0.024
100	0	0.506 $\pm$ 0.199
	1	0.411 $\pm$ 0.190
	4	0.399 $\pm$ 0.264
	20	0.148 $\pm$ 0.124
	100	0.051 $\pm$ 0.036
	400	0.039 $\pm$ 0.031
	2,000	0.035 $\pm$ 0.025

E Per-Replicate Variability in the msprime Scan

649

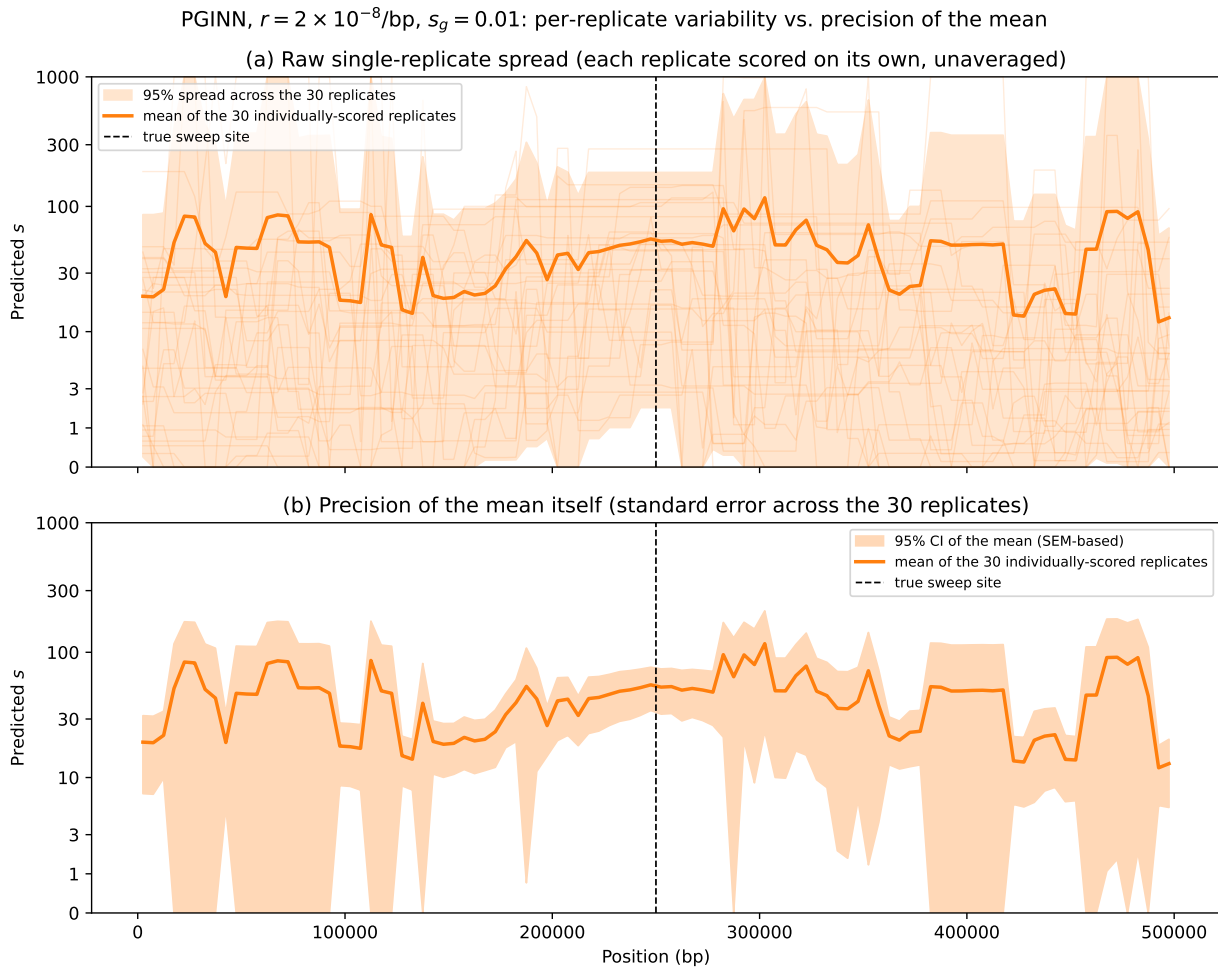


Figure 10: Per-replicate variability underlying Figure 4’s top panel (PGINN, $r=2 \times 10^{-8}/\text{bp}$, $s_g=0.01$): what does the 30-replicate average actually average over? Both panels show s on a $\log(1+s)$ -spaced axis labeled in s units. (a) Each of the 30 replicate simulations scored on its own, rather than averaged into one feature vector per window before scoring. Single replicates are extremely noisy: the median per-window standard deviation across replicates is ≈ 96 against a median predicted s of ≈ 47 ; individual predictions often reach the upper bound of the prior ($s=1000$), and the raw 95% range across replicates has a median half-width of ≈ 105 . (b) The mean of these individually scored replicates with its 95% confidence interval (SEM= $\text{SD}/\sqrt{30}$, median half-width ≈ 34 ; lower bounds below 0 are drawn at 0). Unlike Figure 4, this mean shows no clear peak at the sweep site: the sweep signal there comes from averaging the replicates’ feature vectors before scoring, the 10-loci design the network was trained for, not from averaging per-replicate estimates. A single dataset scored on its own carries little reliable signal, and the sweep recovery in Figure 4 depends on that averaging.

F Illustrative Single-Chromosome Detection Traces

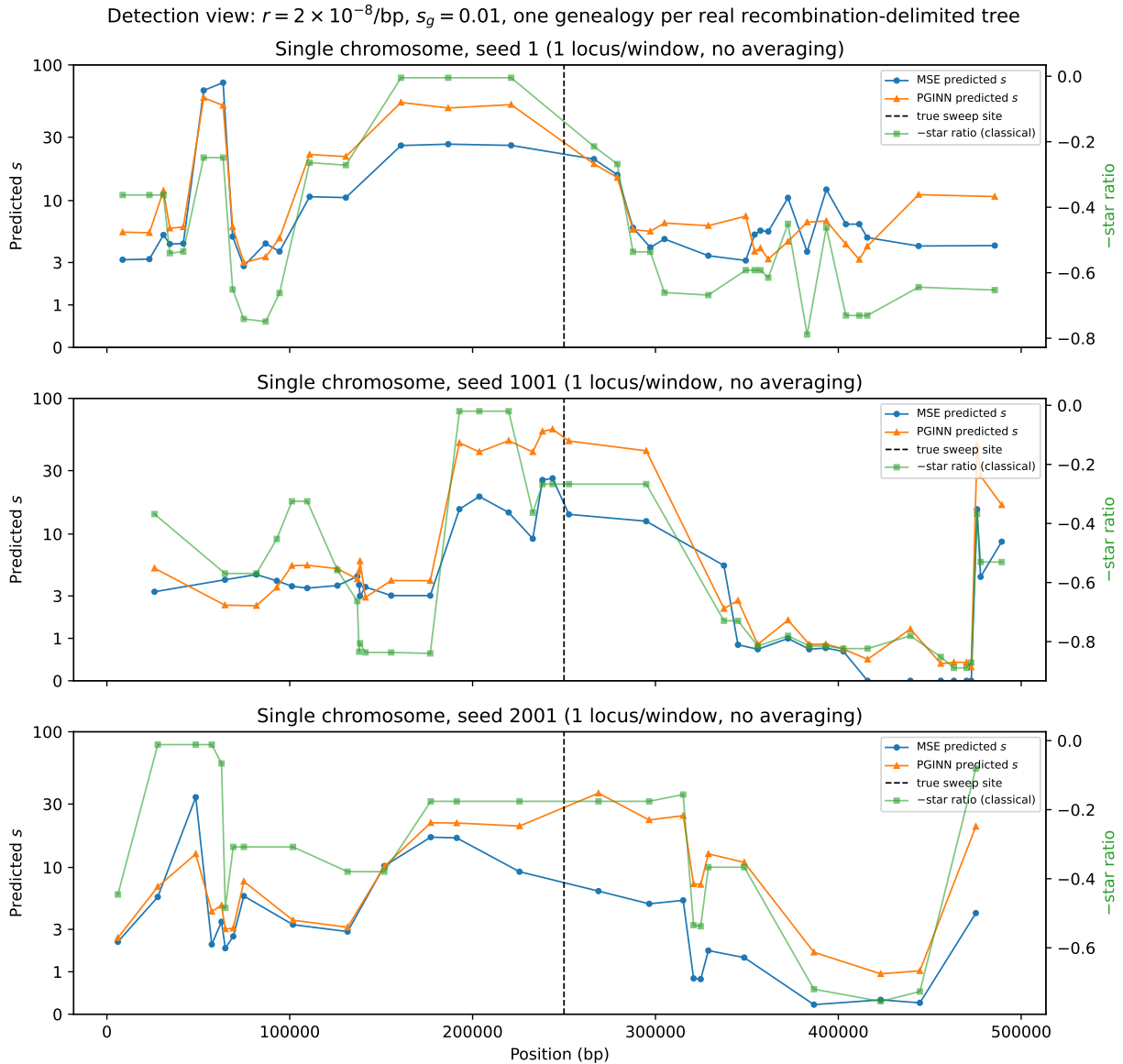


Figure 11: Three illustrative single-chromosome replicates underlying Table 7 and Figure 5: raw, unaveraged predicted s (MSE and PGINN, one tree per position, no cross-tree pooling; left axis $\log(1+s)$ -spaced, labeled in s units) and the classical $-\text{star ratio}$ statistic (right axis), plotted at each chromosome's own recombination-delimited tree boundaries rather than an arbitrary fixed grid. The tree closest to the true sweep site ranks among the top few trees for all three statistics, but not always first: rank 4, 6, and 8 of 34 trees (star ratio, MSE, PGINN) for seed 1 (top), 4, 7, and 4 of 37 for seed 1001 (middle), and 8, 6, and 1 of 25 for seed 2001 (bottom), consistent with Table 7's summary that single chromosomes carry a real but weak signal for every statistic (`code/msprime_scan_detection.py`).

G The McDonald-Kreitman Test at *Adh*, Revisited with More Data

Section 5 notes that McDonald and Kreitman’s (1991) classic test for adaptive protein evolution at *Adh* – an excess of replacement (amino-acid-changing) fixed differences between species relative to replacement polymorphisms within species – no longer reaches significance when their original 11 *D. melanogaster* sequences are replaced with the 176-line DGRP sample used elsewhere in this paper. This appendix gives the full comparison. (`code/mcdonald_kreitman_test.py`).

As the data for the test, we used 176 DGRP *Adh* sequences (Figure 6’s alignment). For our outgroup, we used sequences for *D. simulans* and *D. yakuba*, matching McDonald and Kreitman’s original two-outgroup design, were fetched from NCBI. Each sequence translates to the same single 256-residue *Adh* protein with matching exon lengths (99/405/267 bp) as *D. melanogaster*, confirming orthology and reading frame.

- *D. melanogaster*: RefSeq NT_033779.5:14,615,552–14,618,902 (chr2L, Release 6), the same genomic window used throughout this paper; CDS join(788..886, 952..1356, 1427..1693) relative to that window.
- *D. simulans*: NC_052520.2:14,437,167–14,438,898 (Prin Dsim 3.1 assembly); CDS join(813..911, 979..1383, 1450..1716).
- *D. yakuba*: NC_052527.2:14,861,245–14,863,135 (Prin Dyak Tai18E2 2.1 assembly); CDS join(834..932, 997..1401, 1467..1733).

Following McDonald and Kreitman’s original site-based convention, each candidate site is classified once as synonymous or replacement by substituting the alternate base into the *D. melanogaster* reference codon and comparing the resulting amino acid. A site counts as a *fixed* difference only if it is not segregating within the 176-line DGRP sample; a polymorphic site in *D. melanogaster* is counted under *polymorphic* instead, regardless of whether an outgroup also differs there. This is the specific mechanism by which a larger within-species sample can move a site out of the fixed-difference category that a smaller sample missed.

Table 11: McDonald-Kreitman 2×2 table, our data (DGRP $n=176$, divergence pooled across *D. simulans* and *D. yakuba*) against McDonald and Kreitman’s (1991) original.

	Ours (DGRP $n=176$)		McDonald and Kreitman ($n=6-11$)	
	Repl.	Syn.	Repl.	Syn.
Fixed	8	28	7	17
Polymorphic	4	16	2	42

Fisher’s exact test on our table gives $p=1.0$ (odds ratio ≈ 1.14), against their reported $p=0.006$ (a G -test; Fisher’s exact test on their published counts gives $p=0.007$); the excess of replacement fixations no longer reaches significance. Eight coding sites that differ between *D. melanogaster* and the outgroup are also segregating within the 176-line DGRP sample, and are therefore excluded from the fixed-difference count under the rule above; in a small sample these same sites would have looked like ordinary fixed differences, exactly as they did in 1991. One of the eight is codon 193,

the Fast/Slow site itself (Section 4.8), with 11 sequences it read as a fixed replacement difference; with 176 it is visibly the well-known within-species polymorphism it actually is.

Our polymorphic-synonymous count (16) is *lower* than McDonald and Kreitman’s original (42), which a larger sample alone would not predict. The likely reason is a difference in sampling scheme, not only size: their 11 sequences were drawn from geographically diverse populations across several continents, while the 176-line DGRP sample is drawn from a single North Carolina population. A small, globally diverse sample can carry more scattered old variation per site than a large sample from one population. We therefore read this comparison as showing that McDonald and Kreitman’s original small-sample result does not reproduce under a much larger single-population sample – a caution about small-sample tests for selection generally – rather than as a clean, sample-size-controlled replication, or as evidence against selection having acted on *Adh* at some point in its history (the divergence side of the test, and the broader sweep-vs-balancing-selection question in Section 5, are unaffected by this comparison).

References

- N. H. Barton. The effect of hitch-hiking on neutral genealogies. *Genetical Research*, 72(2):123–133, 1998.
- F. Baumdicker, G. Bisschop, D. Goldstein, G. Gower, A. P. Ragsdale, G. Tsambos, S. Zhu, B. Eldon, E. C. Ellerman, J. G. Galloway, A. L. Gladstein, G. Gorjanc, B. Guo, B. Jeffery, W. W. Kretzschmar, K. Lohse, M. Matschiner, D. Nelson, N. S. Pope, C. D. Quinto-Cortés, M. F. Rodrigues, K. Saunack, T. Sellinger, K. Thornton, H. van Kemenade, A. W. Wohns, Y. Wong, S. Gravel, A. D. Kern, J. Koskela, P. L. Ralph, and J. Kelleher. Efficient ancestry and mutation simulation with msprime 1.0. *Genetics*, 220(3):iyab229, 2022.
- P. Beerli, H. Ashki, S. Mashayekhi, and M. Palczewski. Population divergence time estimation using individual lineage label switching. *G3 Genes—Genomes—Genetics*, 12(6):jkac040, 2022. doi: 10.1093/g3journal/jkac040.
- D. J. Begun, A. J. Betancourt, C. H. Langley, and W. Stephan. Is the Fast/Slow allozyme variation at the *Adh* locus of *Drosophila melanogaster* an ancient balanced polymorphism? *Molecular Biology and Evolution*, 16(12):1816–1819, 1999. doi: 10.1093/oxfordjournals.molbev.a026096.
- J. M. Braverman, R. R. Hudson, N. L. Kaplan, C. H. Langley, and W. Stephan. The hitchhiking effect on the site frequency spectrum of DNA polymorphisms. *Genetics*, 140(2):783–796, 1995.
- B. Charlesworth and D. Charlesworth. Population genetics from 1966 to 2016. *Heredity*, 118:2–9, 2017. doi: 10.1038/hdy.2016.55.
- K. Cranmer, J. Brehmer, and G. Louppe. The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, 117(48):30055–30062, 2020.
- J. Felsenstein, M. K. Kuhner, J. Yamato, and P. Beerli. Likelihoods on coalescents: a Monte Carlo sampling approach to inferring parameters from population samples of molecular data. In F. Seillier-Moisewitsch, editor, *Statistics in Molecular Biology and Genetics*, volume 33 of *IMS Lecture Notes—Monograph Series*, pages 163–185. Institute of Mathematical Statistics, 1999. doi: 10.1214/lnms/1215455552.

- R. C. Griffiths and S. Tavaré. Sampling theory for neutral alleles in a varying environment. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences*, 344(1310): 403–410, 1994. doi: 10.1098/rstb.1994.0079.
- J. Hermisson and P. S. Pennings. Soft sweeps: molecular population genetics of adaptation from standing genetic variation. *Genetics*, 169(4):2335–2352, 2005.
- W. Huang, A. Massouras, Y. Inoue, J. Peiffer, M. Ràmia, A. M. Tarone, L. Turlapati, T. Zichner, D. Zhu, R. F. Lyman, M. M. Magwire, K. Blankenburg, M. A. Carbone, K. Chang, L. L. Ellis, S. Fernandez, Y. Han, G. Highnam, C. E. Hjelman, J. R. Jack, M. Javaid, J. Jayaseelan, D. Kalra, S. Lee, L. Lewis, M. Munidasa, F. Onger, S. Patel, L. Perales, A. Perez, L. Pu, S. M. Rollmann, R. Ruth, N. Saada, C. Warner, A. Williams, Y.-Q. Wu, A. Yamamoto, Y. Zhang, Y. Zhu, R. R. H. Anholt, J. O. Korbel, D. Mittelman, D. M. Muzny, R. A. Gibbs, A. Barbadilla, J. S. Johnston, E. A. Stone, S. Richards, B. Deplancke, and T. F. C. Mackay. Natural variation in genome architecture among 205 *Drosophila melanogaster* Genetic Reference Panel lines. *Genome Research*, 24(7):1193–1208, 2014. doi: 10.1101/gr.171546.113.
- R. R. Hudson. Gene genealogies and the coalescent process. *Oxford Surveys in Evolutionary Biology*, 7:1–44, 1990.
- R. R. Hudson. Generating samples under a Wright–Fisher neutral model of genetic variation. *Bioinformatics*, 18(2):337–338, 2002. doi: 10.1093/bioinformatics/18.2.337.
- R. R. Hudson and N. L. Kaplan. Statistical properties of the number of recombination events in the history of a sample of DNA sequences. *Genetics*, 111(1):147–164, 1985. doi: 10.1093/genetics/111.1.147.
- R. R. Hudson, M. Kreitman, and M. Aguadé. A test of neutral molecular evolution based on nucleotide data. *Genetics*, 116(1):153–159, 1987. doi: 10.1093/genetics/116.1.153.
- N. L. Kaplan, R. R. Hudson, and C. H. Langley. The “hitchhiking effect” revisited. *Genetics*, 123(4):887–899, 1989.
- J. Kelleher, A. M. Etheridge, and G. McVean. Efficient coalescent simulation and genealogical analysis for large sample sizes. *PLOS Computational Biology*, 12(5):1–22, 05 2016. doi: 10.1371/journal.pcbi.1004842. URL <https://doi.org/10.1371/journal.pcbi.1004842>.
- M. Kendall and C. Colijn. Mapping phylogenetic trees to reveal distinct patterns of evolution. *Molecular Biology and Evolution*, 33(10):2735–2743, 2016.
- M. Khodaei, M. Owen, and P. Beerli. Geodesics to characterize the phylogenetic landscape. *PLOS ONE*, 18(6):1–19, 06 2023. doi: 10.1371/journal.pone.0287350. URL <https://doi.org/10.1371/journal.pone.0287350>.
- M. Khodaei, S. V. Edwards, and P. Beerli. Estimating genome-wide phylogenies using probabilistic topic modeling. *Systematic Biology*, 74(5):850–861, 2025. doi: 10.1093/sysbio/syaf015. URL <https://doi.org/10.1093/sysbio/syaf015>.
- J. F. C. Kingman. The coalescent. *Stochastic Processes and their Applications*, 13(3):235–248, 1982.
- M. Kreitman. Nucleotide polymorphism at the alcohol dehydrogenase locus of *Drosophila melanogaster*. *Nature*, 304:412–417, 1983. doi: 10.1038/304412a0.

- 762 M. Kreitman and R. R. Hudson. Inferring the evolutionary histories of the Adh and Adh-dup loci
763 in *Drosophila melanogaster* from patterns of polymorphism and divergence. *Genetics*, 127(3):
764 565–582, 1991. doi: 10.1093/genetics/127.3.565.
- 765 S. Kullback and R. A. Leibler. On information and sufficiency. *Annals of Mathematical Statistics*,
766 22(1):79–86, 1951.
- 767 T. F. C. Mackay, S. Richards, E. A. Stone, A. Barbadilla, J. F. Ayroles, D. Zhu, S. Casillas, Y. Han,
768 M. M. Magwire, J. M. Cridland, M. F. Richardson, R. R. H. Anholt, M. Barrón, C. Bess, K. P.
769 Blankenburg, M. A. Carbone, D. Castellano, L. Chaboub, L. Duncan, Z. Harris, M. Javaid, J. C.
770 Jayaseelan, S. N. Jhangiani, K. W. Jordan, F. Lara, F. Lawrence, S. L. Lee, P. Librado, R. S.
771 Linheiro, R. F. Lyman, A. J. Mackey, M. Munidasa, D. M. Muzny, L. Nazareth, I. Newsham,
772 L. Perales, L.-L. Pu, C. Qu, M. Ràmia, J. G. Reid, S. M. Rollmann, J. Rozas, N. Saada,
773 L. Turlapati, K. C. Worley, Y.-Q. Wu, A. Yamamoto, Y. Zhu, C. M. Bergman, K. R. Thornton,
774 D. Mittelman, and R. A. Gibbs. The *Drosophila melanogaster* Genetic Reference Panel. *Nature*,
775 482:173–178, 2012. doi: 10.1038/nature10811.
- 776 J. Maynard Smith and J. Haigh. The hitch-hiking effect of a favourable gene. *Genetical Research*,
777 23(1):23–35, 1974. doi: 10.1017/S0016672300014634.
- 778 J. H. McDonald and M. Kreitman. Adaptive protein evolution at the Adh locus in *Drosophila*.
779 *Nature*, 351:652–654, 1991. doi: 10.1038/351652a0.
- 780 M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: A deep learn-
781 ing framework for solving forward and inverse problems involving nonlinear partial differential
782 equations. *Journal of Computational Physics*, 378:686–707, 2019.
- 783 D. R. Schrider and A. D. Kern. Supervised machine learning for population genetics: A new
784 paradigm. *Trends in Genetics*, 34(4):301–312, 2018.